

Chapter 8

Cryogenic Aerosols and Supercritical Fluid Cleaning and Surface Conditioning*

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8.1 CRYOGENIC AND SUPERCRITICAL CLEANING TECHNOLOGIES

The importance of precision cleaning in electronic manufacturing areas can hardly be overstated. It is well known that yield and performance of integrated circuit (IC) devices are strongly linked to the cleaning efficiency during their fabrication, as described in Chapter 2. Cleaning has three distinct requirements: (1) removal of inhomogeneous or particulate contaminants, (2) removal of homogeneous contaminants such as organic films and residues of films, including postetch polymers, and (3) removal of ionic or metallic contaminants from surfaces.

Over the years many cleaning technologies, such as aqueous solution, solvent, plasma, vapor phase, laser, dense phase liquid, UV/O₃ (ultraviolet/ozone), cryogenic aerosol, and supercritical carbon dioxide (CO₂), have been used and some are discussed in other chapters of this Handbook. In this chapter, two cleaning methods related to cryogenic aerosols and supercritical CO₂ (SCCO₂) will be discussed. Argon- and/or nitrogen-based cryogenic aerosols are used in IC manufacturing. Recent developments relating to cryogenic aerosols show promise to extend this technology to future IC generations. SCCO₂ applications for IC manufacturing have not yet been introduced into production and remain in the research and development stages. The sections in this chapter will describe the need for these cleaning technologies, their mechanisms, and their specific uses in IC manufacturing.

8.1.1 Need for New Cleaning Technologies

The most demanding industrial cleaning requirements are used in IC manufacturing. The International Technology Roadmap for Semiconductors (ITRS) identifies “killer defects” as 1/2 the DRAM (dynamic random access memory) 1/2 pitch size [1]. Fig. 8.1-1 shows the evolution of technology nodes

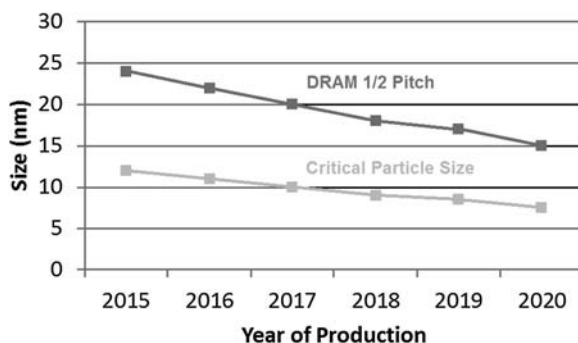


FIGURE 8.1-1 Evolution of integrated device technology and critical particle size [1]. *Compiled by J.M. Lauerhaas from data found in the 2015 ITRS.*

and the minimum critical particle size potentially causing defects in the final device. The current “killer defect” particle size has decreased to 11 nm in 2016 and is expected to further decrease to 7.5 nm in 2020. The continually decreasing defect sizes are driving the need for improved particle removal technologies. Not only must smaller particles be removed but also the removal must occur with minimal material loss.

The removal of submicron particles in front-end-of-line (FEOL) cleaning is conventionally performed with aqueous methods using standard clean 1 (SC-1) consisting of $\text{NH}_4\text{OH}/\text{H}_2\text{O}_2/\text{H}_2\text{O}$ at a typical ratio of 1:1:5, with the addition of megasonic energy or nitrogen atomization. Chapters 1 and 4 provide in-depth discussions of this cleaning process. Particle removal by SC-1 primarily occurs by two mechanisms. The first involves the microetching of the Si surface by the NH_4OH in the SC-1 solution to undercut and lift the particle off the surface. The etching of Si and the subsequent increase in roughness has been described in a wealth of literature [2–9]. The second mechanism by which particle removal occurs is described in the literature [10–13] from the principles of colloid chemistry. The zeta potential of colloids dispersed in a high-pH medium, such as in SC-1, is negative, as is the wafer surface. The resulting ionic double layer repulsion between them prevents the particle lifted off by etching from redepositing back onto the wafer surface. Further discussion of particle behavior in aqueous cleaning is presented in Chapter 3. Typically, megasonic processing is used in conjunction with SC-1 in FEOL to further aid SC-1 chemistry in particle removal. Adding megasonic energy decreases the boundary layer of fluid flow over the wafer surface [14]. The decreased boundary layer thickness results in greater hydrodynamic drag force on particles, otherwise shielded by the liquid layer. Another way that megasonic processing assists SC-1 for particle removal is through acoustic streaming and cavitation, both of which further aid in submicron particle removal. Thus it is the combination of the alkaline chemistry of SC-1 and the megasonic action that results in removal of

submicron particles from Si wafers. However, with the smaller feature geometries on the devices, megasonic energy causes damage to the lines.

The evolution of device manufacturing technology has resulted in the need to remove particles and other contaminants with minimum loss of SiO₂ and polysilicon (polySi) films [1]. Specifically, the ITRS FEOL roadmap states that less than 0.4-Å of SiO₂ and polySi may be lost at the 45-nm technology node and below. Furthermore, the shrinking of the transistor gate lengths with device scaling have made them more fragile and prone to damage; hence new cleaning technologies are being investigated.

As for the back-end-of-line (BEOL), the presence of hydrophobic films such as the low-dielectric constant (low-k) films or chemical mechanical polishing (CMP) hardmasks have introduced challenges to traditional wet and plasma cleaning processes [15,16]. These hydrophobic films have contact angles greater than 70 degrees, as described in Ref. [15], and hence are difficult to clean by conventional aqueous methods. Also of importance for the BEOL cleaning processes is maintaining the bulk film properties, especially the dielectric constant. Conventional cleaning methods using plasma or wet chemistries cause damage to the low-k films and increase the k value. These processes render the films unsuitable for reducing the resistance-capacitance time constant for which they were designed to be used, within the Cu dual damascene integration scheme. In future technology nodes, porous dielectrics (ultralow-k) present new cleaning issues. Penetration of metallic contamination from the Cu or barrier materials, moisture, and other impurities can deleteriously impact device performance and reliability. Methods must be available to effectively remove and prevent intrusion of these moieties. Because low-k materials are susceptible to damage from the plasma etching and stripping processes, new techniques for removing photoresist and antireflection coatings are being developed that prevent damage.

The IC manufacturing industry is continuously improving existing cleaning methods, and simultaneously, examining new cleaning technologies to address the surface conditioning needs. Two such cleaning methods, cryogenic aerosol and SCCO₂, will be described in this chapter. Section 8.2 will focus on cryogenic aerosols and Section 8.3 on SCCO₂. The potential surface conditioning solutions table in the ITRS roadmap [1] includes both cryogenic aerosols and SCCO₂ as addressing the cleaning requirements in IC manufacturing.

8.1.2 Specific Uses of Cryogenic Aerosols and Supercritical Fluids for Surface Conditioning and Cleaning

There are several different areas of cleaning and surface conditioning within the IC manufacturing process where cryogenic aerosols and SCF cleaning are enabling for the future technology nodes. In the FEOL, removal of submicron particles without damage to sensitive metal oxide semiconductor field effect transistor gate electrodes and multigate structures (finFET) [17] is an

important area of application for the future use of cryogenic aerosols. The ITRS FEOL roadmap potential solutions table indicates cryogenic cleaning as a possible cleaning technology for this application. Particle removal using cryogenic cleaning is primarily by momentum transfer, as will be described later in [Section 8.2-1](#) and, hence, does not result in material loss during cleaning. It is also shown in Refs. [18–23] that cryogenic aerosol cleaning is possible without damage to polySi gates, an important requirement for any cleaning technology in FEOL.

In BEOL, the post-CMP cleaning requires removal of slurry particles from hydrophobic films such as low-k films or CMP hardmask materials [15,16]. The predominantly physical mechanism of particle removal using cryogenic cleaning lends itself equally effective to films of low or high surface energies. Another potential application in BEOL is postetch cleaning of dual damascene integration vias and trenches. In this application, the nondamaging cleaning of cryo-aerosols without adverse effects to dielectric films is utilized. Other applications of cryogenic aerosol cleaning in microelectronics manufacturing include pre- and postdeposition, and prelithography cleaning.

Owing to its unusual properties, SCCO_2 enables rapid and penetrating transport of a variety of chemistries into (and out of) narrow structures and complex geometries. Its low surface tension provides stiction-free drying. Consequently, SCCO_2 is a viable method to remove photoresist and postash residues, to repair ash damage and seal pores in low-k films, to remove particles, to clean high aspect ratio features, and to remove moisture and dry narrow features.

8.2 INTRODUCTION TO CRYOGENIC AEROSOLS

8.2.1 Definition of Cryogenic Aerosol and Their Formation

A cryogen (noun) by definition is a liquid, such as liquid N_2 , that boils at a temperature below 110K (-160°C). Cryogenic (adjective) refers in general to low-temperature processes or phenomena. An aerosol is a suspension of solid particles or liquid droplets in a gas. Thus the phrase “cryogenic aerosol” implies a very cold suspension of solid or frozen liquid droplets in a cold gas. The normal boiling points of some of the cryogenic fluids are given in [Table 8.2-1](#) [24].

By definition, the normal boiling point is the temperature, at 1 atm pressure, in which the phase change from a liquid to a gas occurs. Thus by strict definition, CO_2 , which has the highest normal boiling point among the cryogenic fluids list, is not considered as a true cryogen, whereas Ar (argon) and N_2 (nitrogen) are considered cryogens. The details of the thermodynamics of aerosol formation for these systems will be discussed in [Section 8.2-2](#). In general, aerosol formation from these fluids occurs by phase change of the corresponding liquid during a rapid change in pressure and temperature, such as expansion through a nozzle.

TABLE 8.2-1 Normal Boiling Points of Cryogenic Fluids
[24]

Gas	Normal Boiling Point (K)	Normal Boiling Point (°C)
Air	78.7	−194.5
Nitrogen (N ₂)	77.4	−195.8
Oxygen (O ₂)	90.2	−183
Hydrogen (H ₂)	20.3	−252.9
Helium (He)	4.2	−269
Neon (Ne)	27.1	−246.1
Argon (Ar)	87.3	−185.9
Krypton (Kr)	119.9	−153.3
Xenon (Xe)	165.1	−108.1
Methane (CH ₄)	111.7	−161.5
Carbon dioxide (CO ₂)	194.7	−78.5

Compiled by S. Banerjee and L.B. Rothman from data taken from the Handbook of Chemistry and Physics.

8.2.2 Thermodynamic Properties of Gases Suitable for Cryogenic Aerosol Formation

To understand the thermodynamic principles governing aerosol formation of cryogenic fluids, an understanding of the triple point is important [25]. At the triple point condition, all three states of matter (solid, liquid, and gas) coexist with each other in equilibrium. Gibbs phase rule, Eq. (8.2-1), describes the number of degrees of freedom for a given system at equilibrium:

$$F = C - P + 2 \tag{8.2-1}$$

where F is the number of degrees of freedom, C is the number of components in the system, and P is the number of phases in the system. Thus for pure CO₂ ($C = 1$) to exist in equilibrium in all of its three states of matter simultaneously ($P = 3$), Gibbs equation indicates that there are no degrees of freedom. In other words, there is only one temperature and pressure condition, known as the triple point, where CO₂ can coexist as solid, liquid, and gas. At all other temperatures and pressures, one of the phases will evaporate, condense, melt, freeze, or sublime. Table 8.2-2 gives the triple point pressure and temperature of some of the cryogenic fluids listed in Table 8.2-1 [26].

TABLE 8.2-2 Pressure and Temperature Values of Cryogenic Gases at Triple Point [26]

Gas	Triple Point Pressure (atm)	Triple Point Temperature (°C)
Nitrogen (N ₂)	0.12	−210.0
Oxygen (O ₂)	1.44×10^{-3}	−218.8
Hydrogen (H ₂)	0.07	−259.4
Helium (He)	4.2	−269
Neon (Ne)	0.49	−248.6
Argon (Ar)	0.68	−189.4
Krypton (Kr)	0.72	−157.4
Xenon (Xe)	0.80	−111.8
Methane (CH ₄)	0.12	−182.5
Carbon dioxide (CO ₂)	5.11	−56.6

Compiled by S. Banerjee and L.B. Rothman from data published by the Gulf Publishing Company.

The operating conditions (pressure and temperature) of equipment employing cryogenic aerosols are determined by the triple point. In general, the operating condition is lower than the pressure and temperature of the triple point. The reason for this is discussed in [Section 8.2-3](#) where the pressure–temperature diagrams of the CO₂ and Ar/N₂ systems are shown.

Of the different cryogenic fluids mentioned in [Table 8.2-1](#), only CO₂, N₂, and mixtures of Ar and N₂ are used in commercial systems. The other potential cryogenic fluids have not been successful because of various reasons ranging from their potential reactivity with the wafer surface to the difficulty of aerosol formation in terms of pressure and temperature, their unavailability in large quantities, and their cost. [Table 8.2-3](#) gives the approximate cost of some of the cryogenic fluids. It is easy to see that the costs of the rare gases (Ne, Kr, and Xe) are too exorbitant for them to be commercially viable to be used for cleaning. Xenon (Xe), for example, is available only in limited quantities globally, about 48 tons per year, making it prohibitive for wafer cleaning applications requiring continuous supply of the gases.

It is because of these reasons that CO₂, N₂, and Ar/N₂ traditionally are used for cryogenic cleaning. In the subsequent sections the focus will be on CO₂, N₂, and Ar/N₂ cryogenic aerosol processes and systems.

TABLE 8.2-3 Cost of Gases That Can Be Used as Cryogenic Fluids. Updated January 2017

Gas	Price Per Ton (US\$)
Nitrogen (N ₂)	160
Carbon dioxide	110
Hydrogen (H ₂)	7700
Neon (Ne)	300,000
Argon (Ar)	1000
Krypton (Kr)	85,000
Xenon (Xe)	1,850,000

8.2.3 Thermodynamics of CO₂ Aerosol Systems

Fig. 8.2-2 shows the pressure—temperature diagram at the conditions at which CO₂ can exist in a single phase or coexist at equilibrium in multiple phases. The critical point denotes the pressure—temperature condition above which CO₂ has the properties of both a liquid and a gas; this region is known as the SCCO₂ region, which will be described in Section 8.3.

Carbon dioxide gas is stored in cylinders, and under normal conditions, at a pressure of 850 psia (pounds per square inch absolute pressure) and a temperature of about 20°C (ambient or room temperature). Fig. 8.2-2 shows that at

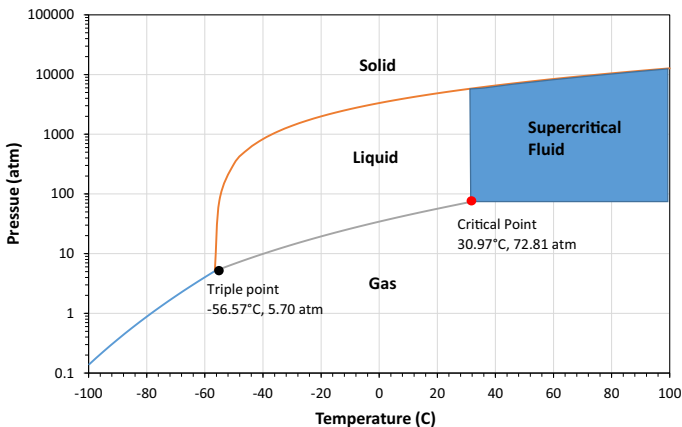


FIGURE 8.2-2 Pressure—temperature diagram of CO₂. The triple point and the critical point are shown with respect to pressure and temperature [26].

the cylinder pressure and temperature, CO_2 coexists in equilibrium in both liquid and gas phases. These cylinders typically have dip tubes to extract liquid CO_2 , which is heavier than the gaseous CO_2 and resides at the bottom of the cylinder. Alternately, the gas from the cylinder head space can also be used for the formation of CO_2 aerosols.

The aerosol in a CO_2 cleaning system is formed by first extracting the fluid from the cylinders, which is then expanded through an orifice of a specially designed nozzle [27,28]. The cleaning chamber is at atmospheric pressure. Thus the fluid undergoes an adiabatic expansion in the nozzle. During this expansion process, the fluid also undergoes Joule–Thomson cooling, resulting in a temperature drop. The combination of pressure and temperature drop of the CO_2 results in a dispersion of solid-phase particles (frozen) in gas-phase CO_2 , which is the aerosol that exits the nozzle. This aerosol stream consisting of frozen CO_2 , also called “snow,” is responsible for contaminant removal that will be described in Section 8.2-5.

Although Fig. 8.2-2 helps one to understand the pressure and temperature changes that the CO_2 fluid undergoes during its transition into an aerosol, it does not shed any insight into the effect of the starting phase on the final phase; the aerosol composition. The pressure–enthalpy diagram of the three phases of CO_2 [26], shown in Fig. 8.2-3, helps explain the dependence of the aerosol composition on the starting fluid phase (i.e., whether it is liquid or gas). As seen in Fig. 8.2-3, at a cylinder pressure of 850 psia and ambient temperature of about 20°C , CO_2 can exist in equilibrium between liquid and vapor phases. Two examples of an aerosol are shown in Fig. 8.2-3, one in which the CO_2 fluid starting phase is liquid and the other in which the starting phase is vapor.

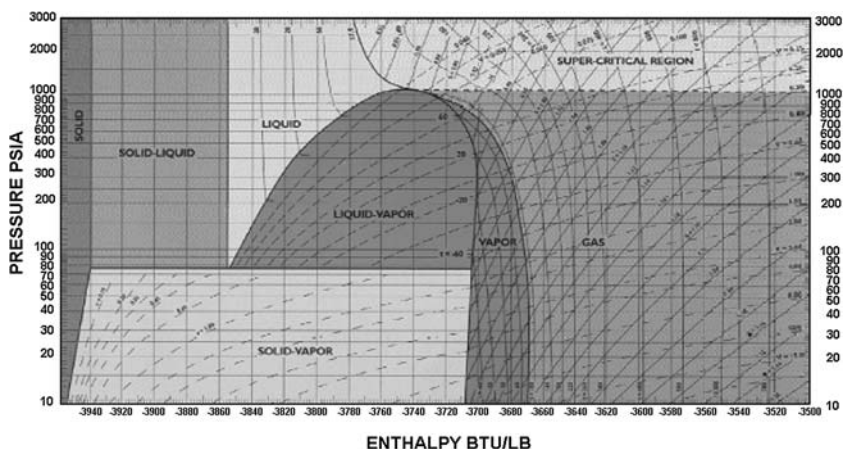


FIGURE 8.2-3 Pressure–enthalpy diagram of CO_2 [26]. Used with permission from Gulf Publishing Company.

Liquid CO_2 is in equilibrium with its vapor phase at 850 psia in Fig. 8.2-3, on the border of the liquid and liquid–vapor regions. Thus starting from liquid phase at 850 psia, as CO_2 expands through the nozzle in an adiabatic and isoenthalpic manner, some of the liquid initially converts to vapor. Furthermore, as the pressure decreases below 5 atm (73.5 psia), the boundary between liquid–vapor and solid–vapor regions are crossed, when a portion of the liquid freezes into particles, whereas the remainder turns into vapor. The percentage of solid in the aerosol starting from liquid CO_2 is about 40%.

Fig. 8.2-3 shows the example of vapor in equilibrium with liquid CO_2 , the liquid–vapor region. As this vapor expands through the nozzle, some portion of this vapor condenses to liquid as the phase crosses into the liquid–vapor region. On further expansion to less than 5 atm (73.5 psia), the phase crosses the solid–vapor boundary. A portion of the liquid freezes into solid, whereas the remainder undergoes a phase change into vapor. The percentage of solid in the aerosol starting from vapor-phase CO_2 is less than 10%.

The pressure–enthalpy diagram of Fig. 8.2-3 therefore demonstrates that the CO_2 aerosol composition and mechanism of formation are dependent on the starting phase of the fluid. This phenomenon provides a means of controlling the areal density of the frozen solid particles in the aerosol plume (i.e., greater fraction of the solids starting from liquid CO_2 and less from gas). This is beneficial when the solid content must be controlled for nondamaging cleaning and surface conditioning process.

8.2.4 Thermodynamics of Ar/ N_2 Cryogenic Aerosol System

The cryogenic aerosol formed from Ar and/or N_2 gas is similar to the phase change that occurs with CO_2 cryogenic aerosol, outlined in Section 8.2-3. The gaseous Ar or Ar/ N_2 is first extracted from the cylinder at 110 psia and at an ambient temperature of about 20°C and then is converted into the liquid phase by precooling. Precooling is achieved by passing the gas or gas mixture through liquid N_2 at -172°C . The precooled Ar or Ar/ N_2 mixture at -172°C and at 77 psia exists in equilibrium between its liquid and gas phases, as shown in the pressure–temperature phase diagram, Fig. 8.2-4. The liquid–gas mixture is then delivered to the process chamber, which is maintained at subatmospheric pressures.

The fluid mixture enters the chamber through a specially designed nozzle where it forms an aerosol of liquid droplets dispersed in the gas phase. As the aerosol exits the nozzle and into the subatmospheric pressure chamber, the liquid droplets freeze during the expansion process through evaporative cooling, thereby generating the cryogenic aerosol inside the process chamber [20]. The phase path during the expansion of the liquid droplets is shown by the dotted arrow in Fig. 8.2-4. The cryoaerosols of frozen, solid particles dispersed in a gas stream are responsible for particle removal. The mechanism of particle removal is discussed in Section 8.2-5.

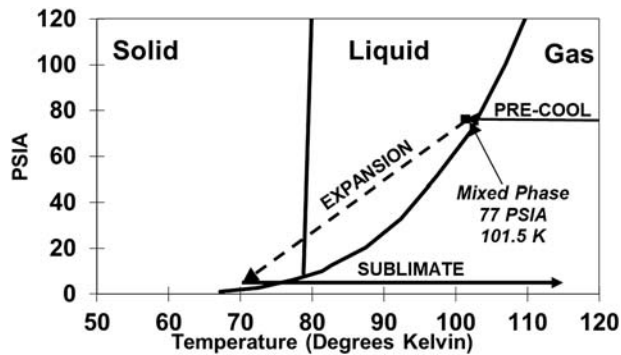


FIGURE 8.2-4 Pressure–temperature diagram of Ar/N₂ system [22] along with the pressure–temperature path taken by the gases to form the cryogenic aerosol. *Courtesy of TEL FSI, Inc.*

Extension of cryogenic aerosols to remove nanometer-sized defects without damage is currently under study [22,23]. Modification to the pre-cooling of the Ar or Ar/N₂ mixtures have resulted in an aerosol that greatly improves particle removal efficiency (PRE) for particles smaller than 40 nm without sacrificing the PRE of larger particles. Damage-free performance of the modified cryogenic aerosol has improved when compared with the previously discussed cryogenic aerosols as well. The main reason for the improved process performance is the different physical properties of the aerosol compared with the previous-generation aerosols. The modified precooling delivers a cryogenically cooled gas opposed to a gas–liquid mixture into a modified dispense nozzle, which results in a smaller (nanometer vs. micrometer) and faster (hundreds of meters per second vs. tens of meters per second) aerosol [29,30]. Data regarding PRE and damage-free processing using this modified aerosol is discussed in Section 8.2-7.

8.2.5 Particle Removal Mechanism in Cryogenic Aerosol Cleaning

This section describes the removal mechanism of inhomogeneous contaminants, such as particles, by momentum transfer using cryogenic aerosols. As discussed in Section 8.2, when a mixture of cryogenically cooled gas and liquid is used to generate the cryogenic aerosol, this results in frozen solid particles entrained in a gas stream. The aerosol coming out of a nozzle is directed at the particles on the wafer surface as shown in the schematic in Fig. 8.2-5. The frozen or cryogenic aerosol particles collide with the contaminant particle thereby imparting force to it, and thus momentum transfer. If the force imparted onto the particle is greater than the force of adhesion, the contaminant particle on the wafer surface is dislodged. The gaseous component of the aerosol then drags the particle away from the wafer

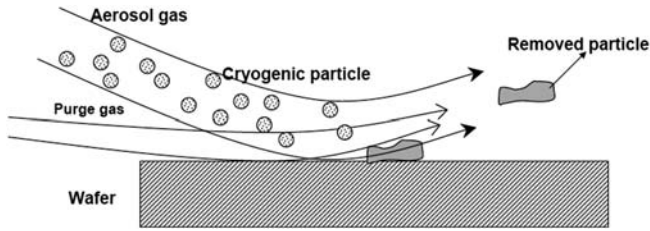


FIGURE 8.2-5 Schematic of inhomogeneous contaminant removal in a cryogenic aerosol cleaning process. *Used with permission from S. Banerjee.*

surface. A purge gas, such as N_2 , flowing parallel to the wafer surface, is used as a curtain to keep the particle from redepositing onto the wafer surface from which it is removed.

The flow of purge gas and the gaseous component of the aerosol stream over the Si wafer surface creates a boundary layer, as described in Chapter 3. The solid cryogenic particles entrained in the gas stream must cross into this boundary layer to arrive at the surface and subsequently remove the contaminant particles [14,19,27,31]. During their flight through the boundary layer, the velocities of the cryogenic particles decrease because of the drag force by the gas within the boundary layer. This decrease in the particle velocity is known as the velocity relaxation time, τ (in μs), defined as:

$$\tau = \frac{2r^2 \rho_p C_c}{9\eta} \quad (8.2-2)$$

where r is the radius of the cryogenic particle, ρ_p is the density of the cryogenic particle (kg/m^3), η is the viscosity of the CO_2 gas (Pas), and C_c is the Cunningham slip correction factor [32] given as:

$$C_c = 1 + \frac{1}{Pr} [6.32 + 2.01 \exp(-0.201Pr)] \quad (8.2-3)$$

where P is the absolute pressure (kPa).

The relaxation time in Eq. (8.2-2) indicates that the velocity of a cryogenic particle crossing the boundary layer will decrease to 36% of the initial velocity if it takes the time equal to one relaxation time. The equation also indicates that the relaxation time increases for larger cryogenic particles implying that they can cross the boundary layer with a larger fraction of their initial velocity. Fig. 8.2-6 shows a calculation of the velocity of cryogenic particles crossing a 50- μm -thick boundary layer at an angle of 45° degrees to the substrate. The figure indicates that particles larger than 2 μm can cross the boundary layer with greater than 90% of their initial velocity [14,18,27].

In general, particle removal is accomplished by the frozen cryogenic particles crossing the boundary layer and colliding with the contaminant particles. There are several types of adhesion forces between the inhomogeneous

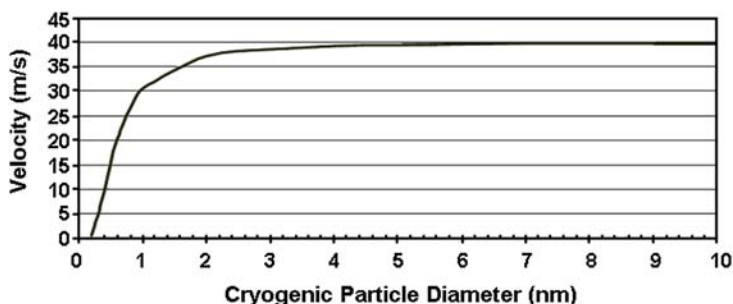


FIGURE 8.2-6 Velocity of cryogenic particles as a function of size upon crossing the boundary layer. *Used with permission from S. Banerjee.*

contaminant particle and the wafer surface. These forces are described in Refs. [13,33] and in Chapter 3. Discussed in this chapter are the different forces of adhesion relevant to a dry cleaning method such as cryogenic aerosol cleaning in which the forces of adhesion exist with air as an intervening medium between the contaminant particle and the wafer.

The relative magnitudes of the adhesion forces of Si particles on Si substrates in air are shown in Fig. 8.2-7 [27,34]. The figure indicates that for submicrometer particles, capillary force is by far the largest force, followed by van der Waals force with deformation, and finally the electronic double layer force. The Coulombic image force is small enough to be neglected for these particles. Although the calculations are for Si particles on Si substrate, the above-mentioned conclusions are also true for a wide variety of material that compose the particles found on Si or SiO₂ films experienced during IC manufacturing. Since the chamber for cryoaerosol cleaning and surface conditioning is generally kept at low humidity (i.e., in CO₂ aerosol cleaning the

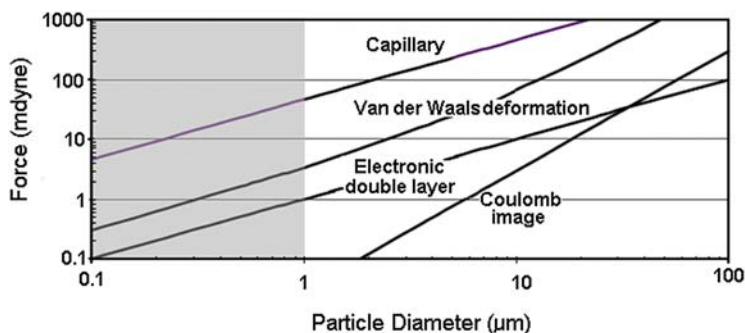


FIGURE 8.2-7 Comparison of forces of adhesion between a Si particle and a wafer in air. The shaded region is the submicrometer particle region and the various forces of adhesion that dominant in that regime. *Used with permission from S. Banerjee.*

humidity is less than -35°C dew point) the capillary and Coulombic image forces do not contribute significantly to the force of adhesion and are neglected. Thus van der Waals is the dominant particle–substrate force of adhesion for this aerosol cleaning method for submicrometer particles on Si wafers.

The forces exchanged between cryogenic and contaminant particles during collision are shown in Fig. 8.2-8 in the two-body collision model. A cryogenic particle is seen crossing the boundary layer before colliding with a contaminant particle adhering to the surface of a wafer. The horizontal and vertical force components are shown in the figure. The boundary layer is considered to be thicker than the contaminant particle diameter, which is true for the submicrometer- and nanometer-sized particles on a wafer surface. The aerosol particle loses velocity by drag force, as indicated in Eq. (8.2-2). The various forces of adhesion and removal are shown in the figure. The horizontal forces (F_T and F_{CT}) help remove the particle, whereas the vertical forces (F_{van} , F_{el} , and F_{CN}) act to hold the particle to the wafer surface. The particle can be removed by either rolling it off the surface or for larger, irregular particles, sliding it off. Whatever the method of particle removal, the force of removal must be greater than the force of adhesion for the particle to be successfully removed.

Another important consideration for particle removal by momentum transfer from impact collision is the angle of impact q , shown in Fig. 8.2-8. The normal collision force F_{CN} is a sine function of the impact angle:

$$F_{\text{CN}} = F_C \sin \theta \quad (8.2-4)$$

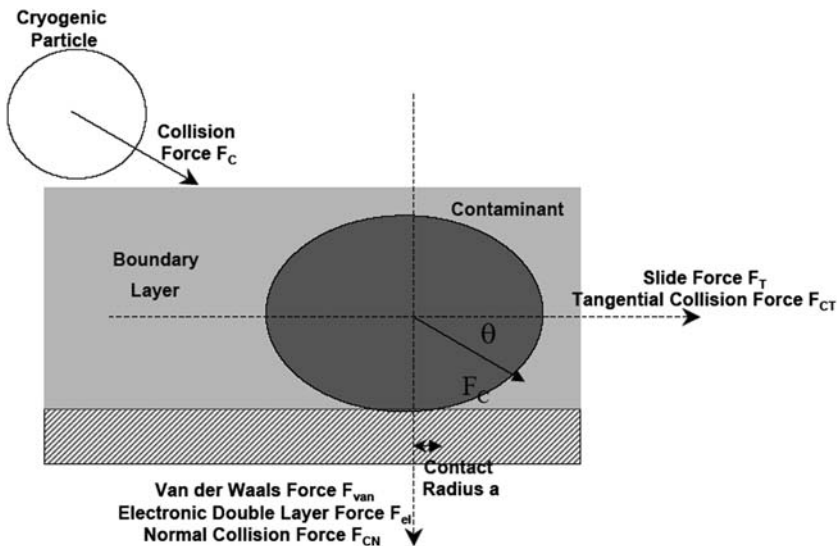


FIGURE 8.2-8 Schematic of particle removal process in cryogenic aerosol cleaning technique. The various forces of removal are outlined. *Used with permission from S. Banerjee.*

whereas the tangential collision force is a cosine function:

$$F_{CT} = F_C \cos \theta \quad (8.2-5)$$

The angle of impact plays an important role in determining the relative magnitude of the tangential collision force versus the normal collision force. The greater the normal collision force, or the higher the impact angle, the less likely it is for the contaminant particle to be removed.

The angle of impact is primarily dictated by the angle of the nozzle with respect to the wafer surface. However, there is yet another factor determining the impact angle: the relative size of the cryogenic and contaminant particles. Fig. 8.2-9 shows how the angle of impact varies with the size of the cryogenic particle colliding with the contaminant particle of a given size. For example, a 0.5- μm cryogenic particle can collide with a 0.1- μm contaminant particle at a minimum angle of 30 degrees. Therefore, in this example, if the nozzle is set at 45 degrees to the wafer surface, as shown by the dotted line, the 0.5- μm cryogenic particles will have no difficulty colliding at angle q of 45 degrees and thus removing the particle. However, the same is not true for a 5.0- μm cryogenic particle colliding with the 0.1- μm contaminant particle, in which the minimum angle of impact is about 75 degrees, and would not remove the particle.

It is therefore important to have a smaller cryogenic aerosol size distribution for effective removal of submicrometer particles below 0.045 μm (45 nm). These smaller cryogenic aerosols have sufficient force for removal of the submicrometer particles but without the damaging aspect of cryogenic cleaning [20–23]. The results of nondamaging cleaning processes will be shown in Section 8.2-7.

8.2.6 Cryogenic Aerosol Generation Description

To create cryogenic aerosol, the process gas for cleaning (either N_2 or a mixture of N_2/Ar) is passed through a liquid N_2 dewar for precooling as

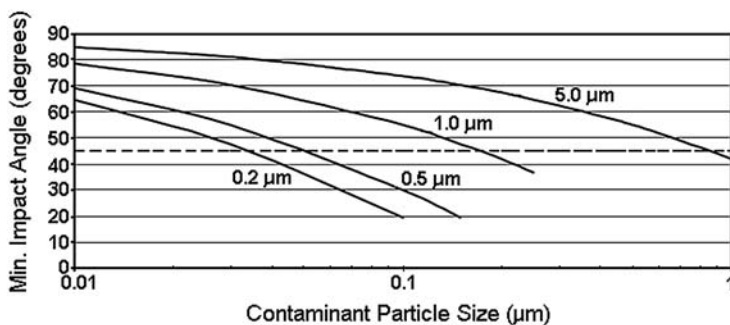


FIGURE 8.2-9 Variation of the impact angle as a function of contaminant particle size for different cryogenic particle diameters. The dotted line denotes the desired 45 degrees impact angle as set by the nozzle. Used with permission from S. Banerjee.

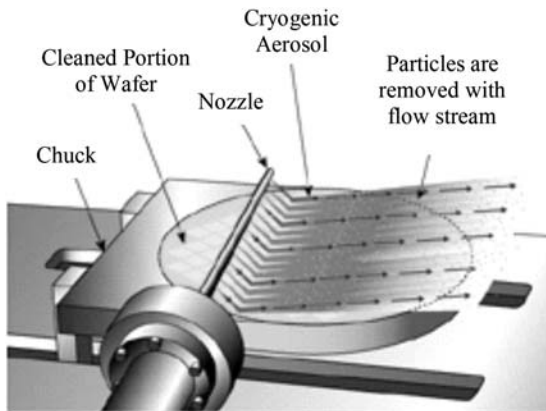


FIGURE 8.2-10 The schematic depicts the nozzle flow pattern and particle transport path during cleaning of a wafer held onto the chuck. *Courtesy of TEL FSI, Inc.*

described in [Section 8.2-4](#). The resulting gas/liquid mixture is then delivered to the process chamber through a nozzle mounted above the substrate to be cleaned, as depicted in [Fig. 8.2-4](#). The nozzle consists of a number of evenly spaced holes capable of processing 300-mm substrates. The process chamber is maintained at a lower pressure than the pressure inside the nozzle. The cryogenic aerosol in the process chamber consists of gas and solid aerosol clusters generated as the gas–liquid mixture is cooled when exiting the nozzle [\[20,30,35,36\]](#). With the proper processing conditions, the solid aerosol clusters impact the substrate, providing the necessary removal force to dislodge the particulates without structural damage ([Fig. 8.2-10](#)). The solid aerosol clusters either sublime or are swept away along with the dislodged contamination by a laminar flow of N_2 gas.

8.2.7 Examples of Cryogenic Cleaning Applications in Microelectronic Manufacturing

[Section 8.1-2](#) described the various applications of cryogenic aerosol cleaning for IC manufacturing processes. In this section, some specific examples and results of those applications are presented.

Researchers compared the N_2 cryogenic aerosol cleaning process to a traditional FEOL SC-1–SC-2 wet bench cleaning process for post-polySi gate etch before silicon nitride (Si_3N_4) spacer deposition. The unsupported polySi gate structures are sensitive to damage. In this analysis, the N_2 cryogenic aerosol process decreased the total defect density without any damage to the unsupported polySi gates. Multiple wafer analysis averaged a 12% defect density reduction for the N_2 cryogenic aerosol process. A split lot comparison between the wet SC-1–SC-2 and N_2 cryogenic process determined that the

absolute product yield was increased by 1.8% with the dry N_2 cryogenic aerosol process [37].

Other researchers have shown, for deposited TiN barrier films, a reduction of the defect density at the second via level after implementation of an Ar/ N_2 cryogenic aerosol cleaning process [38–40]. The scanning electron microscope (SEM) images in Fig. 8.2-11A show two representative defects

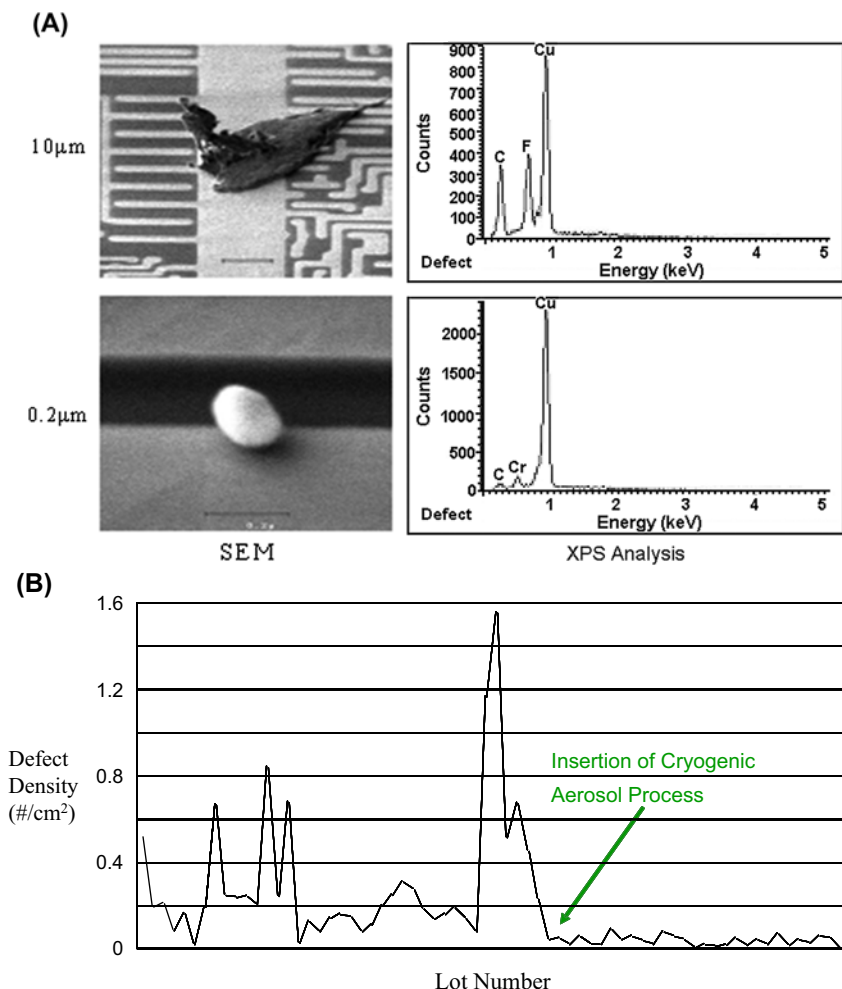


FIGURE 8.2-11 (A) Back-end-of-line (BEOL) cleaning results with a cryoaerosol cleaning system showing typical defects removed and their representative X-ray photoelectron spectroscopy (XPS) spectra showing the compositional analysis [38,39]. (B) BEOL cleaning showing resulting defect densities after insertion of Ar/ N_2 cryogenic aerosol for post-Cu chemical mechanical polishing process [39]. Courtesy of TEL FSI, Inc. Reprinted from *Micro Magazine*. Copyright 1999 and 2002, Tom Cheyney, Los Angeles, CA. Used with permission.

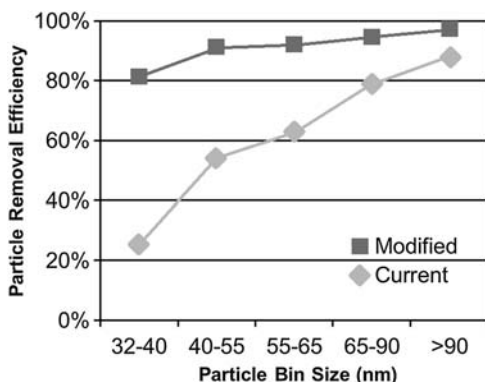


FIGURE 8.2-12 Particle removal efficiency (PRE) comparison of the modified Ar/N₂ cryogenic aerosol compared with the current-generation Ar/N₂ cryogenic aerosol. PRE is increased across all particle bin sizes with the modified cryogenic aerosol [22]. *Courtesy of TEL FSI, Inc.*

of size 10 and 0.2 μm . The X-ray photoelectron spectroscopy (XPS) analysis, shown in Fig. 8.2-11B, indicates that both particles contain copper (Cu). The Ar/N₂ cryogenic aerosol cleaning process successfully removed both particles.

Fig. 8.2-12 shows the particle removal efficiency with respect to particle size. The chart shows a significant reduction in defects after the insertion of the cryogenic aerosol cleaning process [22]. The modified cryogenic aerosol discussed in Section 8.2-4 improves PRE when compared with the current-generation cryogenic aerosols. Fig. 8.2-12 contains a PRE comparison of silicon wafers contaminated with dry-deposited silicon nitride (Si_xN_y) particles that are aged for 36 h before cleaning. The standard cryogenic aerosol process is compared with the modified cryogenic aerosol process, which removes particles across all sizes, especially for smaller particle sizes less than 55 nm [22]. These particles are measured with a commercially available system, as outline in Chapter 11.

Another PRE study further demonstrates nanometer-sized particle removal with modified cryogenic aerosol. Silica (SiO_x) particle of sizes 40 and 30 nm were wet deposited onto silicon wafers and allowed to age for 6 h before cleaning. Particles of both sizes are removed efficiently; 99% PRE for 40 nm and 96% for 30 nm, as shown in Fig. 8.2-13 [22].

Damage-free performance is enhanced with the modified cryogenic aerosol; for example, polySi lines with a 22 nm critical dimension and 200 nm height were processed without damage by the modified cryogenic aerosol. The current cryogenic aerosol, although able to process larger critical dimension and lower height features without damage, was not able to process the features in Fig. 8.2-14 without damage [22].

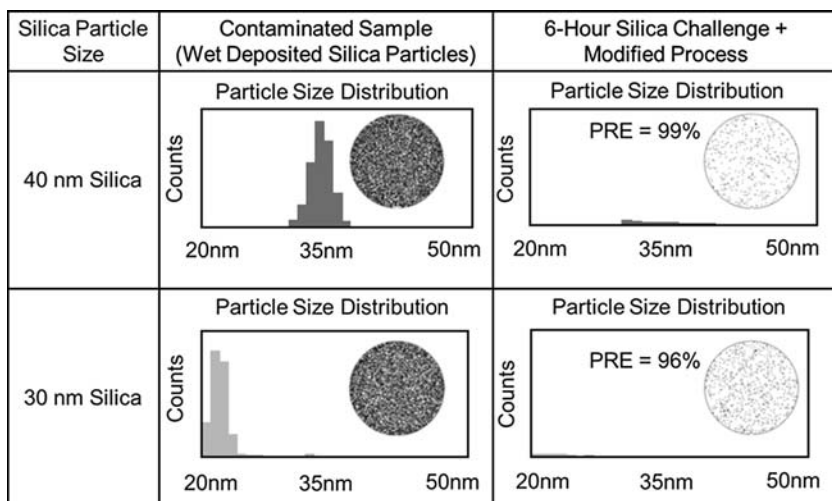


FIGURE 8.2-13 Removal of 40- and 30-nm sized nanoparticle with the Ar/N₂ modified aerosol. High removal efficiency is achieved for both particle sizes [22]. *PRE*, particle removal efficiency. *Courtesy of TEL FSI, Inc.*

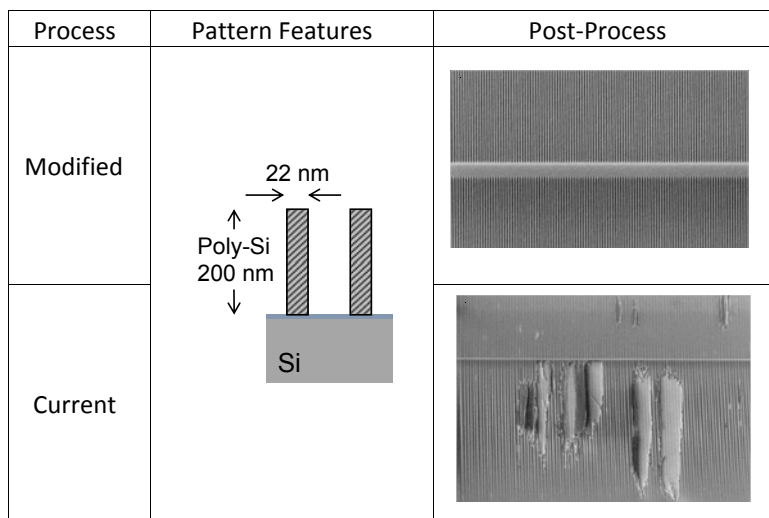


FIGURE 8.2-14 Damage-free cleaning of 22 nm width and 200 nm height polySi lines shown on a 300-mm wafer with the Ar/N₂-modified cryogenic aerosol process. The current cryogenic aerosol process damages the same features [22]. *Courtesy of TEL FSI, Inc.*

8.2.8 Summary of Cryogenic Cleaning

Cryogenic aerosol is a dry, nonaqueous cleaning method. The particle removal is primarily by physical momentum transfer and hence does not etch or oxidize

surface films that may occur with conventional wet cleaning. This surface cleaning method addresses a number of challenges posed by conventional wet and plasma cleaning for IC manufacturing and is an integral part of the ITRS roadmap for advanced cleans. For example, the technology has been implemented for N_2 and Ar/N_2 cryoaerosols to address particle removal without damage to FEOL transistor gates or the high aspect ratio patterns in the BEOL.

8.3 INTRODUCTION TO SUPERCRITICAL AND DENSIFIED FLUID CLEANING

The discovery of SCF dates back to 1879 when Hanney and Hogarth reported [42] on the ability of a supercritical fluid to dissolve low-vapor pressure solid materials. They found that gases could be good solvents under supercritical conditions and that their dissolving power was pressure dependent. At the time there was great controversy over the findings that a “gas” could dissolve solid materials. Many researchers, for quite a number of years, have studied the solubility phenomena of SCF. However, it was not until the mid-1960s that the use of SCF gained attention for extraction processes to eliminate residual solvents in food products. In 1978, a plant was built in Bremen, Germany, for the decaffeination of coffee using $SCCO_2$. Since that time, the uses of SCF have been demonstrated for a variety of applications, including pharmaceuticals [43], textile processing [44], dry cleaning [45], polymerizations [46], coatings [47], food extractions [48], chromatography [49], and precision cleaning [50].

In the early 1990s the Joint Association for the Advancement of Supercritical Technology, a research consortium consisting of industry, university, and national laboratory partners, was formed to help facilitate and stimulate the use of SCF for the replacement of traditional solvents used for cleaning processes. Environmental concerns and the need to eliminate ozone (O_3)-depleting chlorofluorocarbons used for cleaning processes motivated this research. The focus of this initial work was on precision cleaning of parts such as electronic assemblies, optical and laser components, electromechanical elements, ceramics, plastics, and machined metals [51]. The cleanliness requirements for these cleaning operations are far less stringent than those for semiconductor applications.

In the mid-1990s an interest in cleaning semiconductor devices with SCF began. This was partially driven by a desire to reduce H_2O and chemical consumption for cleaning semiconductor wafers. In addition to these environmental concerns, there were technical challenges for cleaning the increasingly complex semiconductor structures. As device dimensions approach the nanoscale, aqueous-based cleaning processes were hampered because of high surface tension and capillary forces, for example, effective penetration into high-aspect ratio trenches and via structures is challenging. An outgrowth of this was cooperative research and development work between

Los Alamos National Laboratory (LANL) and industrial partners, which culminated in the development of the SCORR (SCCO₂ for resist removal) system [52]. Research and development work at IBM [53] and the formation of two supercritical fluid equipment companies brought interest to the potential use of this technology for semiconductor processes.

Although the development of supercritical fluid processes for semiconductor device manufacturing show feasibility, implementation of these methods is limited. For many new technologies, entry into semiconductor manufacturing requires an application in which the novel method outperforms the current methods and the new method overcomes any implementation complications. Although supercritical fluid processes show great promise for residue removal, low-k repair, and etching, the technology was unable to overcome these hurdles. Van Hoeymissen and Malhouitre presented several additional issues that thwarted the introduction of SCCO₂ into IC fabrication: contamination from the hardware, high-pressure equipment reliability, and process uniformity [54]. A review of SCCO₂ in semiconductor device fabrication by Romang and Watkins suggests that, although the capabilities of supercritical methods continue to grow, the introduction of these processes may depend on the successful implementation in non-IC application where the technologies can mature and lower the risk for IC manufacturers [55].

8.3.1 Definition of Supercritical Fluids

SCF are compounds that exist above their critical temperature and critical pressure. SCF are considered a fourth state of matter where the material is neither a liquid nor a gas, but has the properties of both. In the supercritical state, the fluid has the diffusivity of a gas with the solvency of a liquid and has zero surface tension. These are very desirable properties for cleaning and drying densely packed, high-aspect ratio structures in advanced semiconductor devices.

8.3.2 Concept of Supercritical Fluid Cleaning

In addition to the efficacy of the process, the introduction of supercritical critical fluids into IC device production is dependent on several issues: cost of ownership, infrastructure requirements, and environmental health and safety issues. In terms of choosing the appropriate supercritical fluid, CO₂ meets each of these concerns.

8.3.2.1 Properties of Gases Suitable for Supercritical Fluids

In Table 8.3-4, the critical parameters of several fluids are noted. As expected, noble and inert gases have parameters that require relatively low pressure and low temperature, when compared with the other compounds, whereas organic compounds have more severe critical requirements, relatively high temperatures

TABLE 8.3-4 Critical Parameters for a Range of Gases and Liquids [56]

	Critical Temperature (°C)	Critical Pressure (Psia)	Critical Density (g/cm ³)
Helium (He)	−268	33	0.070
Nitrogen (N ₂)	−147	493	0.313
Argon (Ar)	−122	705	0.536
Carbon dioxide (CO ₂)	31	1070	0.468
Chlorodifluoromethane (CHClF ₂)	96	724	0.524
Ammonia (NH ₃)	132	1644	0.225
Hexane (C ₆ H ₁₄)	235	440	0.233
Cyclohexane (C ₆ H ₁₂)	280	591	0.273
Benzene (C ₆ H ₆)	289	710	0.309
Toluene (C ₇ H ₈)	321	614	0.290
Water (H ₂ O)	374	3200	0.322

Compiled by S. Banerjee and L.B. Rothman from Data Obtained from the National Institute of Standards and Technology.

and pressures. Carbon dioxide stands out because the supercritical point is at 31°C and 1070 psia (11.5 atm), a temperature that is easy to achieve, although the pressure is higher than other possible SCF.

In the construction of production-worthy tools, the integrity of the valves and seals are important when working at high temperatures and pressures; therefore, the materials used must withstand either high pressure or temperature, but not necessarily both. For example, many polymer seals can withstand high pressure, but at high temperatures stability becomes an issue. As the critical pressures and temperatures are merely the thresholds, solubility of cosolvents may demand much higher temperatures. Carbon dioxide has reasonable solubility for many cosolvents and has both a relatively low supercritical pressure and temperature, and will not challenge the temperature or pressure limits of many equipment components as other supercritical solvents may.

8.3.2.2 Pressure–Temperature Diagrams

The accommodation of cosolvents or reactants in a supercritical medium demands both solubility and effective transport to the reaction surface. As seen in Table 8.3-4, the critical density of SCCO₂ ($\rho_{\text{critical}} = 0.468 \text{ g/cm}^3$) is less than

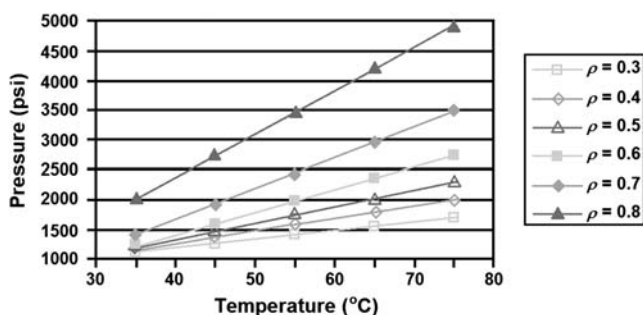


FIGURE 8.3-15 The pressure and temperature dependence for supercritical CO₂ (densities, ρ , are in g/cm³) [56]. Compiled from data obtained from the National Institute of Standards and Technology.

that of liquid CO₂ ($\rho_{\text{liquid}} = 0.86 \text{ g/cm}^3$ at 10°C); consequently, mass transport of a reactant is quite rapid, even for low concentrations. Furthermore, the solubility of many cosolvents into CO₂ is dependent on the density of the CO₂. Because of this physical parameter, the amount of reactant available for a reaction can be controlled. The temperature of the supercritical fluid will affect reaction kinetics due to the Arrhenius equation; however, higher temperatures will require an increased pressure to maintain the same density; increasing the temperature from 40 to 60°C and maintaining a density of 0.6 g/cm³ requires approximately a 50% increase in pressure (1403–2160 psia). Fig. 8.3-15 describes the pressure–temperature dependence for the density of SCCO₂, and demonstrates that balancing kinetics and mass transport requires flexible reactor capability.

8.3.3 Overview of Supercritical Fluid Cleaning Systems

An entire book could be filled describing the design of systems for supercritical fluid cleaning processes; however, in this section the content will be limited to an overview of the basic supercritical fluid system and its major components. A generic system diagram is provided in Fig. 8.3-16. The system has the following components:

1. The CO₂ supply is in the form of gas cylinders or bulk supply tanks.
2. A high-pressure pump is used to bring the CO₂ above the critical pressure and to the desired working pressure.
3. The cofluid reservoir and pump add cosolvents or other additives to the CO₂.
4. A heater is used to raise the temperature above the critical temperature and to the desired working temperature.
5. The processing vessel is provided where the wafer is placed and the process occurs.

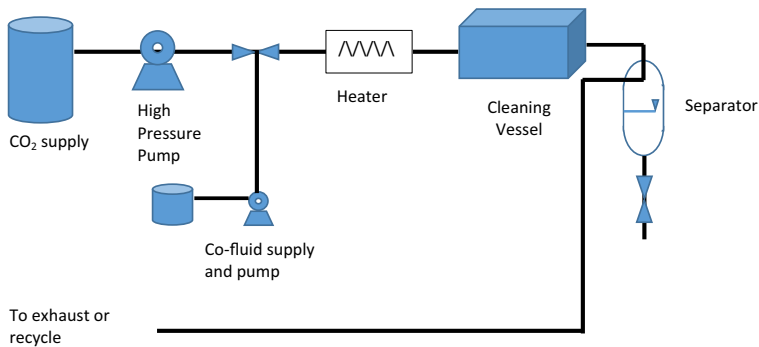


FIGURE 8.3-16 Diagram of generic supercritical fluid system. *Used with permission R.F. Reidy.*

6. A separator is used where, through the expansion, the CO₂ is gasified and bulk contaminants precipitate out of solution and are collected for disposal.
7. All valves, piping, and safety devices are rated for high-pressure usage.
8. The optional recycling system is used to purify the CO₂ and to enable reuse within the system.

Since the design of this vessel is for high-pressure use, it is tested, certified, and code-stamped by a representative of the American Society of Mechanical Engineers. Safety devices include pressure relief valves and rupture disks.

This generic system configuration is similar to systems used for most supercritical fluid processes. The major differences for the application of SCF to the processing of semiconductor wafers are the system requirements for contamination control. Semiconductor devices are much more sensitive to particulate and molecular contamination than even food or pharmaceutical applications. Particle diameters of 20 nm can result in killer defects on a semiconductor device [1]. Molecular contamination from oils or lubricants in the system can also become defects on a semiconductor wafer. In the food and pharmaceutical industry, this problem is avoided by making the system lubricants from food-grade materials.

There are two concepts for system configuration that have developed. One is to have a bulk system with a large high-pressure pump remote from the cleaning vessels, as shown in Fig. 8.3-17. An alternative approach is to use a smaller high-pressure pump contained within the cleaning system footprint. In this approach, the exposure to high pressure is contained within a small area.

8.3.4 Applications Using Supercritical Fluid Cleaning

8.3.4.1 Photoresist and Polymer Removal

Photoresist must be removed from semiconductor wafers numerous times during the IC fabrication process. The ease or difficulty of removal depends upon the processes that the photoresist was subjected to, such as heat

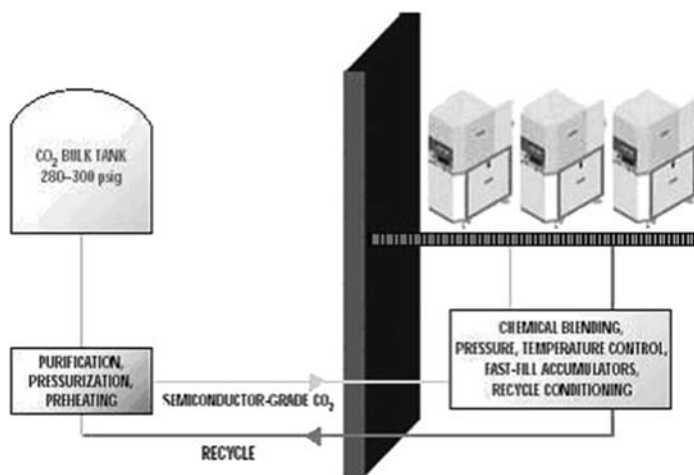


FIGURE 8.3-17 Schematic diagram of a production scale supercritical CO_2 system [57]. Reprinted from *Micro Magazine*. Copyright 2004, Tom Cheyney, Los Angeles, CA. Used with permission.

treatments, plasma etching, or ion implantation, before the removal process. Many investigators [58–66] have demonstrated the use of SCF to remove photoresist under a variety of conditions. The early work on the SCORR process by LANL indicated that the photoresist swells by the diffusion of CO_2 into the polymer matrix and that the cosolvent helps to degrade the polymer and break bonds at the surface. Fig. 8.3-18A shows an example of swollen photoresist in the midst of processing. Fig. 8.3-18B shows the complete removal of the photoresist layer.

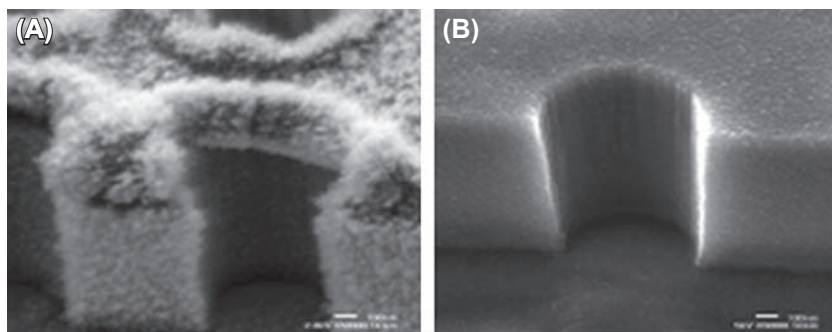


FIGURE 8.3-18 Photoresist removal (A) midway through the supercritical CO_2 (SCCO_2) process and (B) after exposure, showing complete photoresist removal. Cosolvents added to the SCCO_2 using the SCORR process [59]. Used with permission from L.B. Rothman, presented at the Green & Sustainable Chemistry Conference, Royal Society of Chemistry.

The mechanisms for removing photoresist using SCCO₂ processing differ from a typical aqueous chemistry approach. Instead of dissolving the photoresist, the SCCO₂ can be used to swell the polymer and then debond it from the surface. Photoresist that has been subjected to a plasma etch has a “polymeric crust” along the edge of the pattern and top surface, which is more difficult to remove than the underlying material. Therefore the use of this debonding approach can be quite effective.

Examining photoresist samples midway through the process can provide details on the mechanisms. Fig. 8.3-19 shows that the photoresist is swollen and attacked where the edges of the pattern have begun. Since the SCCO₂ can penetrate through the “crust” it can attack the underlying photoresist and actually start to extract it. Continuing with the processing and introducing pulses of rapid depressurization and then repressurization indicates that the photoresist debonds from the surface and is lifted off in sheets, as shown in Fig. 8.3-20.

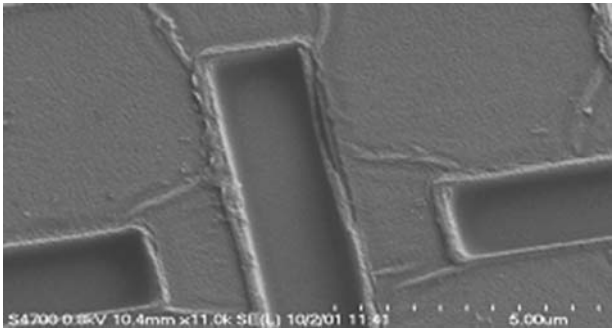


FIGURE 8.3-19 Penetration of supercritical CO₂ and cosolvent into photoresist, midway in the process. *L.B. Rothman, R.J. Robey, M.K. Ali, D.J. Mount, Supercritical fluid processes for semiconductor device fabrication, in: Advanced Semiconductor Materials Conference (ASMC) Proceedings. Copyright (2002) IEEE. Used with permission.*

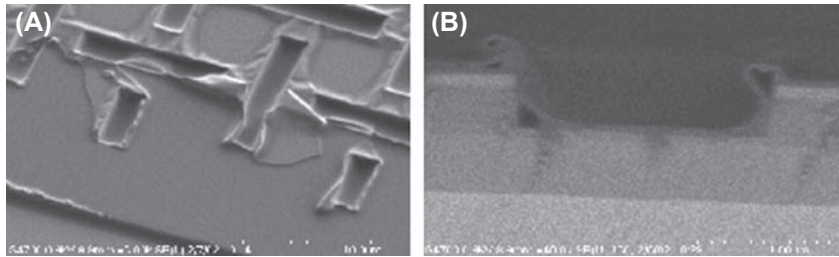


FIGURE 8.3-20 After depressurization and repressurization of the supercritical CO₂ to debond the photoresist; (A) top view and (B) cross-section [58]. *Used with permission from L.B. Rothman, presented at the NSF/SRC Annual Meeting 2003.*

The additional requirement is that there be sufficient fluid flow to remove the debonded photoresist. If the fluid flow is insufficient, the debonded photoresist remains on the wafer surface. The similarity exists with respect to the thinning of the boundary layer as observed with aqueous cleaning, discussed in Chapter 3.

For most chemical reactions, increasing the temperature will accelerate the process. However, for supercritical fluid processes, there are competing forces so that higher temperatures cannot be assumed to yield better results. At higher temperatures the density of the fluid is lower, which could result in lower solubility. The glass transition temperature is the temperature above which a polymeric substance softens and becomes rubbery and viscous. Studies by Sawan et al. [68] have shown that CO_2 at high pressure can plasticize polymers and cause significant reductions in the glass transition temperature. This has been observed and is shown in Fig. 8.3-21; when the temperature is increased the photoresist melts and flows instead of being removed.

Two factors that affect a polymer's ability to solubilize CO_2 are the crystallinity and the cross-linking in the polymer. The presence of crystallinity in a polymer will inhibit the sorption of CO_2 . Highly cross-linked polymers solubilize less CO_2 than those with less cross-linking. The sorption of CO_2 results in swelling of the polymer and is further enhanced if the glass transition temperature is exceeded. Additional disruption of the polymer can occur through interactions with the solubilized CO_2 and/or cosolvent. Upon depressurization, the solubilized CO_2 begins to expand as the vessel pressure decreases. This expansion of the gas and the reduced hydrostatic pressure

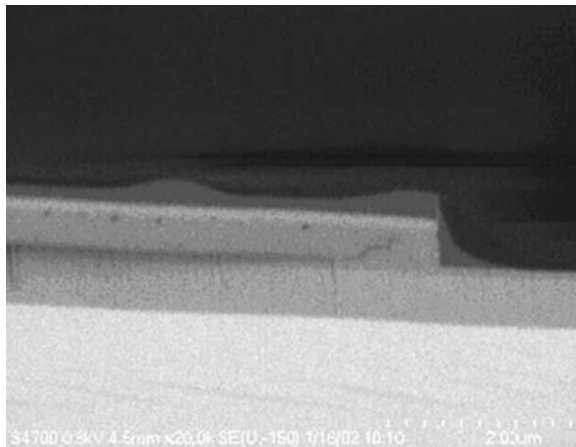


FIGURE 8.3-21 Melt and flow of photoresist after exceeding glass transition temperature [67]. L.B. Rothman, R.J. Robey, M.K. Ali, D.J. Mount, *Supercritical fluid processes for semiconductor device fabrication*, in: *Advanced Semiconductor Materials Conference (ASMC) Proceedings*. Copyright (2002) IEEE. Used with permission.

exerted on the polymer often results in a temporarily enhanced polymer swelling effect. Before the polymer/polymer interactions can be reestablished, stress cracks, bubbles, or complete debonding of the polymer film can occur as the CO_2 rapidly desorbs. For the photoresist to be completely removed, three mechanisms must be combined to constitute a process: (1) CO_2 and cosolvent at the appropriate temperature and pressure swell the polymer, (2) rapid decompression debond and delaminate the polymer from the surface, and, finally, (3) fluid flow removes the polymer off the wafer surface.

An alternative approach for photoresist removal using SCCO_2 is to employ a cosolvent system that will dissolve the “crust” layer and underlying photoresist. The Hansen solubility parameters are used as a guide to cosolvent selection for photoresist removal [60,69]. The cosolvent system may include more than one solvent. Since a polar solution does not easily dissolve in nonpolar SCCO_2 , the addition of an intermediary solvent with polar and nonpolar groups that are soluble in SCCO_2 can be used.

Korzenski et al. [61] found that the addition of polar cosolvents to SCCO_2 is effective for low-dose ion-implanted photoresist but not for medium or higher doses (doses are discussed in Chapter 7). For dissolution of the high-dose ion-implanted photoresist, chemical additives are necessary. The use of a reduction or a fluorination mechanism to form SCCO_2 -soluble fluorides or hydrides results in complete removal of the photoresist by a dissolution process. Before and after results are shown in Fig. 8.3-22.

Levitin et al. [62] investigated the reactions between tetramethylammonium hydroxide and CO_2 for removing photoresist and postetch residues. Studies demonstrated that it is the bicarbonate salt acting as the active ingredient for cleaning. The use of tetramethylammonium bicarbonate as an effective additive to the SCCO_2 system is also studied as a function of phase

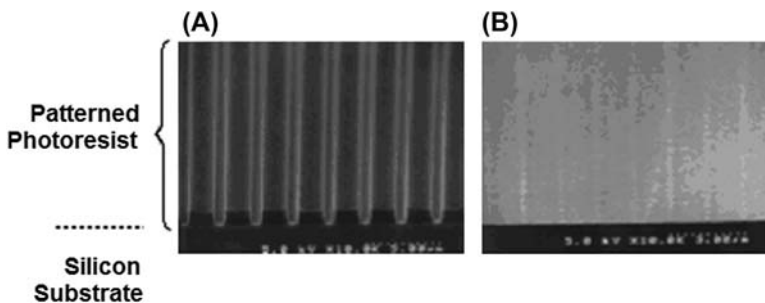


FIGURE 8.3-22 Scanning electron microscopy image of the (A) ion-implanted deep ultraviolet photoresist control sample and (B) completely stripped sample after processing using supercritical CO_2 with a chemical additive formulation. Reproduced by permission of ECS—The Electrochemical Society from M.B. Korzenski, C. Xu, T.H. Baum, K. Saga, H. Kuniyasu, T. Hattori, in: *Eighth International Symposium on Cleaning Technology in Semiconductor Device Manufacturing*.

behavior. The researchers found that single-phase mixtures result in more efficient cleaning of postetch residues than the two-phase mixtures. Tanaka et al. developed several fluorinated quaternary salts that dissolve photoresist polymers in SCCO₂ [70]. Kim et al. showed that ultrasonic agitation in SCCO₂ enhances the removal of implant-hardened resist [71]. In related work, SCCO₂ microemulsions (surfactants and dimethyl sulfoxide) coupled with magnetic agitation effectively removes implanted resists [72].

8.3.4.2 Post-plasma Stripping Residue Removal

The approaches for removing post-plasma stripping residue are similar to photoresist removal. One concern has been the large amounts of cosolvents and active chemistries that are added to SCCO₂ to support ionic dissociation and solvation of poststrip residues. Another approach is the use of H₂O-in-CO₂ microemulsions [57,63]. The surfactants added to CO₂ have a polar head group that point inward and a CO₂-philic tail that point outward, forming what is known as a reverse micelle. These micelles are believed to remove process residues. The polar domain in the microemulsion serve three functions: first to provide solvation for the medium; second to remove water-soluble polar contaminants; and third to deliver the active chemistry to remove the remaining residues. Fig. 8.3-23 shows a schematic of a surfactant design.

Experimental results indicated that 80% of poststripping residues is removed with pure H₂O-in-CO₂ microemulsions; however, for complete 100% removal of residues the addition of active chemistry is required [63]. SEM and transmission electron microscopy (TEM) images are shown in Fig. 8.3-24.

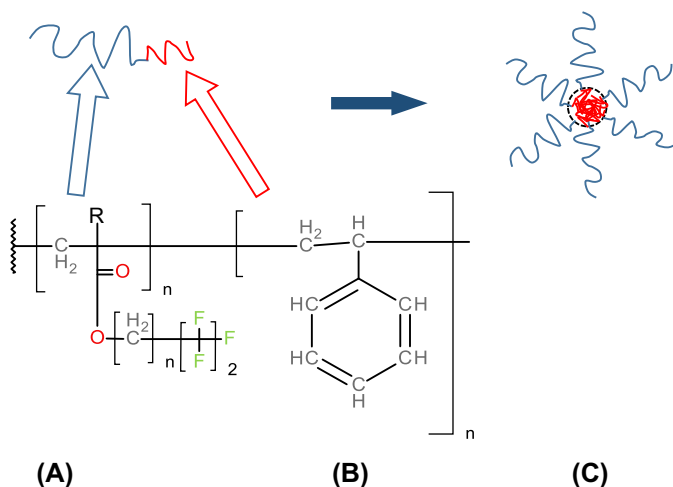


FIGURE 8.3-23 Design of surfactants and formation of micelles in supercritical CO₂: (A) CO₂-philic structure, (B) CO₂-phobic structure, and (C) micelles [57]. Reprinted from *Micro Magazine*. Copyright 2004, Tom Cheyney, Los Angeles, CA. Used with permission.

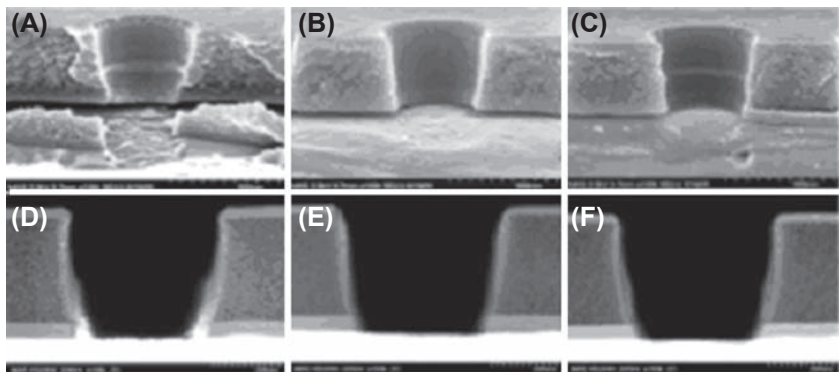
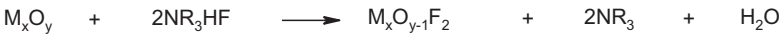


FIGURE 8.3-24 Scanning electron microscopy (top row) and transmission electron microscopy images (bottom row) of posttrench feature of a low-k film on a coated Si wafer; (A) and (D) show the wafer before cleaning; (B) and (E) show the wafer after processing with H₂O-in-CO₂ micro-emulsion containing active chemistry; (C) and (F) show aqueous cleaned samples micelles [57]. Reprinted from *Micro Magazine*. Copyright 2004, Tom Cheyney, Los Angeles, CA. Used with permission.

For Al postetch residue removal, Korzenski et al. [61] have developed a multicomponent system consisting of an organic cosolvent, an etchant, a hydroxyl-containing chelator for postetch residue removal along with a corrosion inhibitor to protect the Al and poly-Si surfaces. The dissolution mechanism can involve fluorination or chelation of the metal residue as depicted in Fig. 8.3-25.

The theory in both cases is to make a post-SCCO₂ reaction product that is more soluble in the SCCO₂ cosolvent solution than the postetch residue. Before and after results are shown in SEM images in Fig. 8.3-26.

Scheme 1: Fluorination



Scheme 2: Chelation

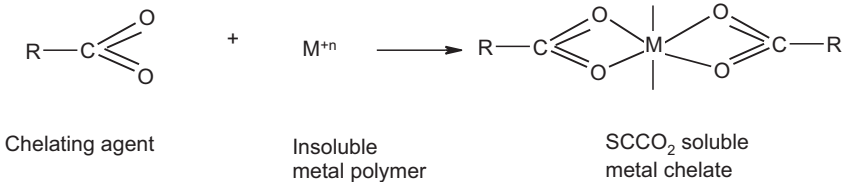


FIGURE 8.3-25 Scheme 1: metal ion converted to an oxyfluoride. Scheme 2: metal ion chelated by organic group [61]. Reproduced by permission of ECS—The Electrochemical Society from M.B. Korzenski, C. Xu, T.H. Baum, K. Saga, H. Kuniyasu, T. Hattori, in: *Eighth International Symposium on Cleaning Technology in Semiconductor Device Manufacturing*.

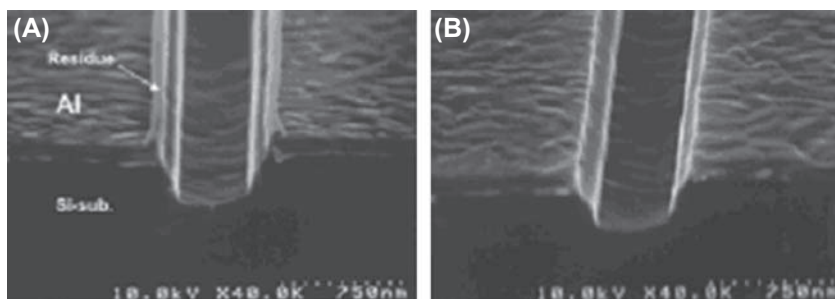


FIGURE 8.3-26 Scanning electron microscopy images of samples with postetch residues (A) before cleaning and (B) after supercritical CO_2 cleaning [61]. *Reproduced by permission of ECS—The Electrochemical Society from M.B. Korzenski, C. Xu, T.H. Baum, K. Saga, H. Kuniyasu, T. Hattori, in: Eighth International Symposium on Cleaning Technology in Semiconductor Device Manufacturing.*

8.3.4.3 Particle Removal

Similar approaches to swelling and debonding can be used for organic particle removal. The dynamics of particle removal in SCCO_2 were first described by Chitanvis et al. [73]. However, for inorganic particles different methods are demonstrated. Korzenski et al. [74] have described the use of chemical additives that include a combination of etchant, surface passivator, and fluorinated surfactant in SCCO_2 to remove Si_3N_4 particles from Si or SiO_2 surfaces. Particles are removed by an undercut mechanism whereby the particle/surface adhesion layer is attacked by the etchant as described in Chapter 3. To minimize the etching of the surface, a passivator is added. It is a fine balance between chemistry and concentration along with processing conditions, such as temperature and pressure, to remove the particle but not overetch the underlying surface. Fig. 8.3-27 shows particle removal results. As pressure increases, the density of the SCCO_2 solution increases, which improves the solubility of the chemical additives, thus the assumption that PRE improves with increasing pressure.

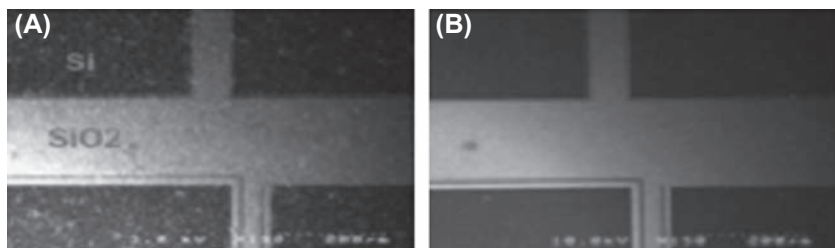


FIGURE 8.3-27 Optical microscope images of (A) a patterned wafer with Si_3N_4 particles on the surface and (B) the wafer processed in supercritical CO_2 with chemical additives [74]. *Used with permission from Scitec Publications.*

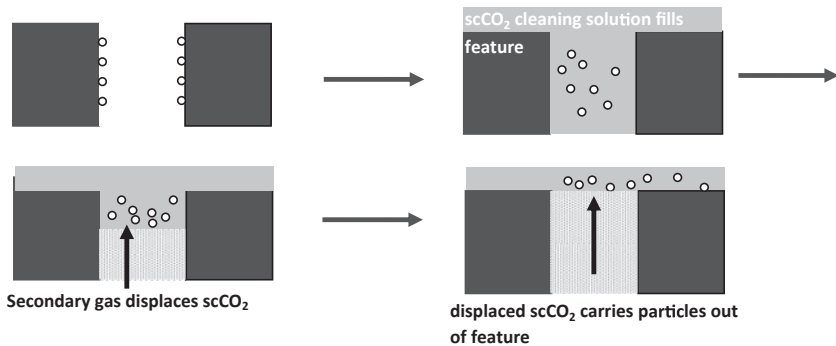


FIGURE 8.3-28 A depiction of a supercritical fluid cleaning process [67,68]—particles are removed from the feature by cleaning chemistries in the supercritical CO_2 (SCCO_2), the secondary gas displaces the SCCO_2 and the entrained particles. *Used with permission of R.F. Reidy.*

Another method for particle removal has been developed employing the use of a secondary gas into SCCO_2 [75,76]. Because the secondary gas is insoluble in the supercritical phase, this method also permits incorporation of additional cleaning elements within the SCCO_2 . After a requisite resident time on the wafer, the SCCO_2 cleaning solution is displaced from patterned structures because of pressure from the secondary gas. As it is “pushed out” of features by the secondary gas, the SCCO_2 carries particles and soluble species with it. The process is depicted in Fig. 8.3-28.

8.3.5 Dielectric Repair Applications

Although H_2O has limited solubility in CO_2 , liquid and supercritical CO_2 can effectively remove moisture from porous low- k films. Because the dielectric constants of insulators used for interlayer dielectrics (ILD) are extremely sensitive to absorbed H_2O ($k = 78$), prevention and/or removal of H_2O are critical to device performance and reliability. Many silica-based low- k films are inherently hydrophobic because of inclusion of methyl groups (CH_3-) within the $\text{Si}-\text{O}$ network. However, plasma processes typically used for IC manufacture, such as etching and photoresist stripping (as described in Chapter 7), remove these hydrophobic C-moieties to a significant depth, which are replaced with hydrophilic silanol groups ($\text{Si}-\text{OH}$). Dynamic secondary ion mass spectroscopy profiles indicate that the C-depleted regions may be as thick as $400 \text{ \AA}$ [77]. The sample geometry and plasma species employed in the plasma stripping process affects the type and concentration of the resulting silanol varieties: isolated or unhindered-rotator, vicinal or hindered-rotator (e.g., hydrogen-bonded), and geminal, as shown in Fig. 8.3-29.

Owing to their polar nature, these hydroxyl ($-\text{OH}$) groups will impact the dielectric constant of the ILD films directly by increasing the film orientational polarization, and indirectly through the absorption of moisture. As stated

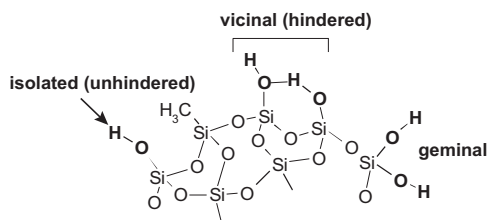


FIGURE 8.3-29 Silanol ($\text{Si}-\text{OH}$) species created during photoresist stripping processes of porous low-k films. The resulting silanol are isolated or unhindered-rotator, vicinal or hindered-rotator, and geminal. Note the hydrogen bonding that is also observed. *Used with permission of R.F. Reidy.*

previously, pure SCCO_2 , despite limited solubility, can remove significant amounts of H_2O in a dynamic flow system. Cosolvents (e.g., alcohols) serve to increase the solubility of H_2O in SCCO_2 , enhancing H_2O extraction. Even at modest porosities, uncapped low-k films rapidly uptake moisture in wafer fab environments. To block the absorption or reabsorption of H_2O , silylating agents are dissolved in SCCO_2 . These agents include, hexamethyldisilazane (HMDS) [78–80], trimethylchlorosilane (TMCS) [81,82], trialkylchlorosilanes [83], dialkyldichlorosilanes [84] alkyltrichlorosilanes [84], and monoalkyl (long chain) dimethylchlorosilanes [85,86]. In addition, the hygroscopic nature of the silylating agent permits the reaction and removal of some surface-adsorbed H_2O . TMCS, HMDS, or other alkylsilanes can react with silanols to form trialkylsilyl groups, repairing the native surface hydrophobicity of the films as shown in Figs. 8.3-30 and 8.3-31. Porous $\text{Si}-\text{O}-\text{C}$ films and organosilicate glasses (p-OSG) are employed as low-k dielectrics in copper interconnect structures. These porous films are deposited by chemical vapor deposition (CVD); consequently, they have some limitation in their mechanical properties. However, the addition of porogens during deposition can generate very

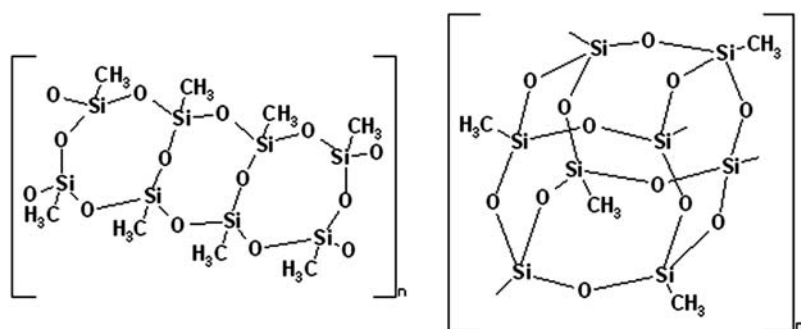


FIGURE 8.3-30 Hexamethyldisilazane silylation reaction of surface silanols in porous low-k. *Used with permission from R.F. Reidy.*

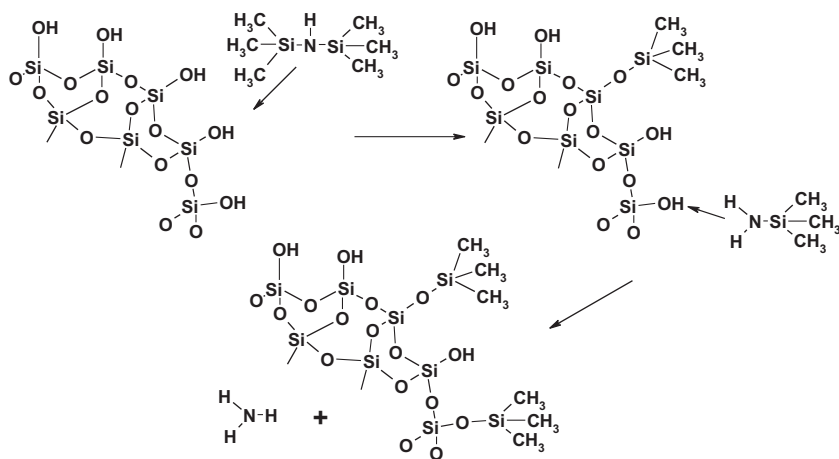


FIGURE 8.3-31 A comparison of various plasma stripping processes on porous low-k and the subsequent repair [92]. Reprinted with permission from B.P. Gorman, R.A. Orozco-Teran, Z. Zhang, P.D. Matz, D.W. Mueller, R.F. Reidy, *J. Vac. Sci. Technol. B22* (2004), 1210. Copyright 2004, American Institute of Physics.

small pore sizes commensurate with current scaling needs. CVD deposition also allows graded densities and controlled fraction of hydrophobic groups.

The reason for the preference of silylation reaction with isolated or vicinal silanols of the porous low-k is unclear. There is some evidence that vicinal species react preferentially with liquid HMDS [87]; however, this finding is not universally accepted [88]. Gun'ko et al. suggest that HMDS preferentially reacts with free silanol and subsequently one silanol at each geminal site. The remaining geminal silanol reacts much slower while hydrogen-bonded silanols show extremely limited reactions. Gun'ko attributes limited HMDS reaction with silanols to steric effects in fumed silica particles. Based on Fourier transform infrared spectroscopy results, Xie and Muscat reported that HMDS reacted with a large fraction of the hydrogen-bonded vicinal species [79].

The relative reactivity is in part a function of the post-plasma stripping process surface condition. For blanket films (such as those used by Xie and Muscat) or for trench bottoms, the plasma-generated, high-kinetic energy species impinge directly on the film, resulting in high removal rates of methyl groups and a concomitantly high proportion of silanols. As the fraction of surface silanols increases, the probability for the occurrence of adjacent (or vicinal) silanols likewise increases. The surface conditions of the silica species described by Xie and Gun'ko differ considerably and neither represents the plasma exposure experienced by trench or via sidewalls; therefore, it is difficult to determine the relative reactivity of HMDS on plasma damaged in patterned device wafers.

Although the reaction moiety preferences associated with SCCO_2 silylation reaction are not well understood, exposures to silylating agents increase the film contact angle (i.e., hydrophobicity) at relatively low concentrations, 1 vol % (volume percent), and short reaction times, 1–2 min [78]. As discussed previously, the contact angles after silylation are affected by the nature of the plasma damage. The contact angles for O_2 - and H_2 -based plasma stripping processes do not change significantly as a function of HMDS reaction time. A hydrogen-plasma stripped low-k has a very low contact angle, but the hydrophobicity increases (within 1–2 min) to the values of the preprocessed low-k. For O_2 -based plasma stripping conditions, the repaired contact angle does not reach the unprocessed value, but does attain its maximum value within 1–2 min [89]. The lower hydrophobicity of repaired low-k after an O_2 -based compared with H_2 -based plasma stripping processes is likely due to the unavailability of surface sites for silylation. If a preponderance of vicinal silanols is formed during the O_2 -based stripping process, then bridged O-species ($\text{Si}-\text{O}-\text{Si}$) that would not be susceptible to silylation also form. Reaction of vicinal species to form bridged O-species occurs during bakes of 400°C . Plasma-stripped porous low-k samples that are baked before silylation exhibit lower contact angles, suggesting that the vicinal species are no longer available for silylation and fewer trimethylsilyl groups are produced.

Pressure and temperature do not show a significant effect on silylation, although carbamate ($-\text{NH}(\text{C}=\text{O})\text{O}-$) formation can occur under high-pressure conditions. Porous low-k reactions with HMDS under two different conditions, $40^\circ\text{C}/1400$ psia and $60^\circ\text{C}/2700$ psia, have shown quite similar results [77]. The lower-pressure data indicate a saturation of silylation within 1 min of exposure at a contact angle of about 63 degrees. At 2700 psia, the contact angles reach approximately 50 degrees within 1 min, and slowly increase as a function of time. The difference in these contact angles is likely due to sample conditions. The 2700-psia samples were handled and measured within a clean room environment, whereas the 1400-psia samples were not. Because the cleanroom samples would have a limited number of surface particles, the additional airborne particles on the 1400-psia silylated wafer could very likely increase the contact angle. The gradual increase in contact angle for the 2700-psia samples was studied using XPS to examine changes in the surface C species concentration. XPS spectra for unprocessed, O_2 -plasma stripped, and SCCO_2 -HMDS-repaired low-k show a significant loss of C species (i.e., methyl groups) after the stripping process and an increase in C species after SCCO_2 -HMDS repair [77].

Dynamic secondary ion mass spectroscopy results suggest the depth of the silylation repair is limited [77]. Increased pressure and temperature of the supercritical medium increases the penetration of HMDS into the porous film; however, SIMS results do not show a significant difference between the C-depth profiles of the plasma stripped compared with the repaired samples, at depths greater than 20-Å below the surface. Furthermore, no differences are

observed in the C concentration profile as a function of HMDS reaction time [77]. The failure of HMDS to penetrate deeply into the films is likely due to initial surface silylation reactions that block further diffusion of HMDS into pores and, thus, prevent reaction with silanols below the surface. Although C-moiety replacement is limited within the film bulk, surface silylation performs a key function—the prevention of moisture adsorption. For example, a plasma stripping process can increase porous low-k dielectric constant from a value of 2.2 to a value of 3.5, indicating significant depletion of C. After $\text{SCCO}_2/\text{HMDS}$ treatment, the dielectric constant can be repaired to approximately 2.5, although the degree of repair is variable depending on the plasma stripping process.

Nonpolar methyl group replacement by a polar silanol group leads to a slight increase in k , but this is dwarfed by the increased polarization due to absorbed H_2O [77]; for plasma-stripped films the increase of dielectric constant is due to H_2O absorption rather than the replacement of methyl groups with silanol groups. Using D_2O as a moisture uptake marker, Vinogradova et al. showed that oxidizing plasma stripping processes absorb a limited amount of moisture, whereas reducing processes are much more susceptible to water uptake [90,91].

SCCO_2 repair of the plasma-damaged low-k also prevents film loss during subsequent wet etching steps [92]. The plasma-induced loss of hydrophobic methyl functionalities leaves the-MSQ (methylsilsequioxane) vulnerable to aqueous attack; in the case of H_2/N_2 plasma-stripped MSQ as much as 1000 Å of the MSQ film may be removed during etch in 200:1 $\text{H}_2\text{O}/\text{HF}$. The depth and the nature of the plasma damage determine the extent of attack. For example, O_2 -based plasma-stripped samples incurred significantly thinner thickness losses than the H_2/N_2 -based plasma-stripped MSQ. Fig. 8.3-32 shows that an

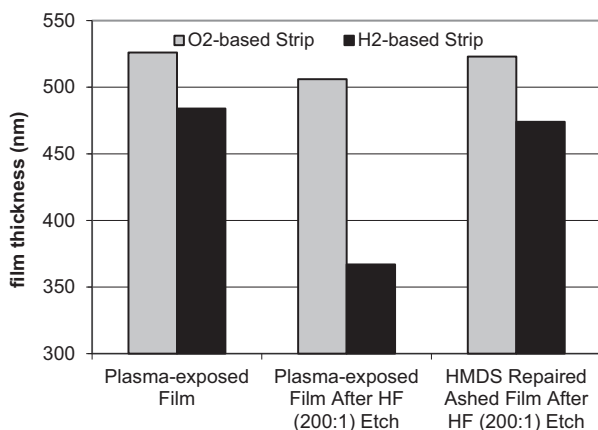


FIGURE 8.3-32 A comparison between supercritical CO_2 cleaning process (cosolvent and hexamethyldisilazane [HMDS]) and a baseline semiaqueous cleaning process using comb-serpent line-to-line devices of increasing pitch (smallest pitch = comb-serpent test device for a spin-on porous low-k after O_2 ash) [77]. Used with permission from Trans-Tec.

H₂-based plasma-stripped MSQ suffers severe film loss, whereas an SCCO₂/HMDS-repaired film does not show any loss of material. However, an O₂-based plasma-stripped film showed less thickness loss after HF etching than H₂-stripped films, and the SCCO₂/HMDS-treated films did not show any thickness losses following HF etching [92].

Because of the ease of direct measurement of dielectric constants, studies of plasma-damaged and the subsequently repaired blanket MSQ films prevail in the literature. In Fig. 8.3-33, a dual-damascene comb-serpentine MSQ-based test structure is employed to compare k-value repair efficacy of SCCO₂ with dissolved HMDS relative to that of typical post-plasma-stripped F-based wet cleaning process [77]. For the smallest pitch test structures, the normalized comb-serpentine line-to-line capacitance for SCCO₂/HMDS samples decrease by 18% relative to those samples receiving the wet-clean. For the largest pitch case, the capacitance is reduced by only 7%. Because the plasma-damaged region between the metal lines is approximately constant, the proportion of the capacitance due to the damage is much larger for the smaller pitch sample, as shown in Fig. 8.3-34. Consequently, the repair of the dielectric constant becomes more critical as the pitch decreases.

SCCO₂ has been shown to remove by-products caused during film deposition and low-molecular weight species such as organic contaminants. Furthermore, it has been demonstrated that the infrared transmittance of a porous low-k increases after SCCO₂ exposure [93]. Line-to-line test structures

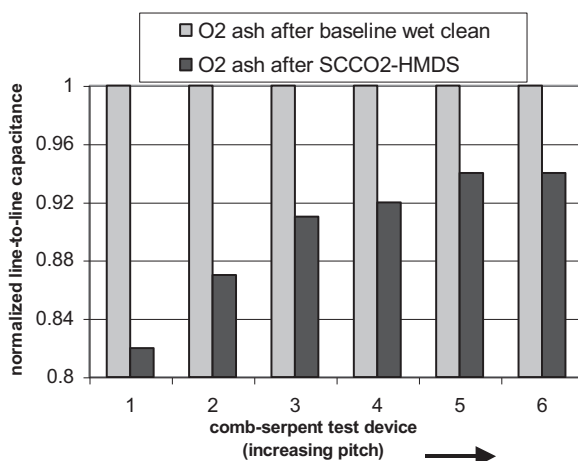


FIGURE 8.3-33 Schematic representation of the effect of plasma-damage dielectric thickness and the impact of subsequent repair on measured line-to-line capacitance. As the pitch decreases, the capacitance of the damaged region has a greater contribution to the total capacitance. Consequently, repair of the damaged region becomes more critical at smaller pitches. HMDS, hexamethyldisilazane; SCCO₂, supercritical CO₂. Used with permission from R.F. Reidy.

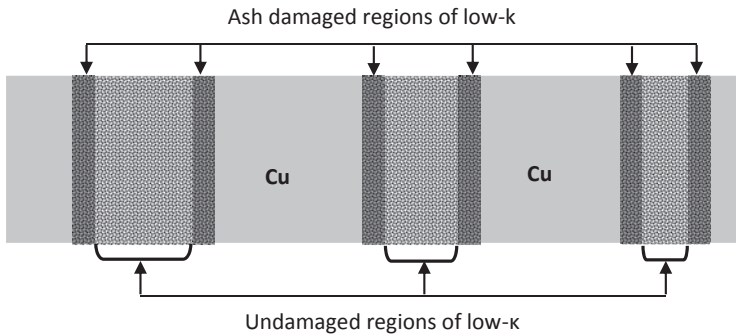


FIGURE 8.3-34 Titanium penetration into porous low-k film after reaction with tetrakis-dimethylamido titanium. (A) O_2 -based plasma processed methylsilsesquioxane (MSQ) film and (B) O_2 -based plasma processed MSQ film after treatment with $SCCO_2$ /hexamethyldisilazane [77]. Used with permission from Trans-Tec.

of a porous CVD low-k exhibited significantly elevated dielectric breakdown field voltage from $SCCO_2$ exposure compared with baseline clean processes [77], as demonstrated at device operation and accelerated lifetime temperatures.

8.3.6 Supercritical Fluid Metal Removal, Etching, and Surface Conditioning

8.3.6.1 Removal of Copper

Several groups have examined the removal of Cu residues and CuO_x from wafer substrates using SCF [80,94,95]. Previously, $SCCO_2$ with β -ketonate additives have been used to extract Cu from filter paper [96]. In experiments on semiconductor materials, both fluorinated [72,94] and nonfluorinated [95] β -ketonates were employed. Using these chemistries, cupric oxide (CuO) has a low etching rate and must be removed selectively [95]. Xie et al. observed that cuprous oxide (Cu_2O) must be oxidized to ensure removal and employed rapid pressure release in combination with chemical attack to delaminate the CuO [94]. The addition of sulfonic acid and diketones to $SCCO_2$ also has been shown to be an effective postetch cleaning process [97].

8.3.6.2 Etching of Silica

$SCCO_2$ with F-based additives is shown to effectively etch silica-based films. Dilute solutions of 0.05 vol% HF/ethanol in $SCCO_2$ are capable of removing a 300-nm blanket porous low-k film in less than 60 s [98]. Owing to limited pore penetration, this rate is slowed if the etchant is diluted in liquid CO_2 . Jones et al. employed 129–260 μM solutions of HF/pyridine (70:30 wt%) in $SCCO_2$ to remove thermally grown SiO_2 [99]. SiO_2 removal rates exceeded 20 $\text{\AA}/\text{min}$ over 20 min. The etching rates are dependent on temperature; 75°C removal rates were 3.5 times faster than those at 35°C. Also using dilute F-based

etchants (10 vol%) in SCCO_2 , Kaga et al. compared the effects of different organic cosolvents on SiO_2 etching rates [100], which significantly affected the rates (800–5000 Å/min). In addition, incorporation of the etchant in SCCO_2 enhances rates over aqueous etchants with the same concentration of F^- . Xie et al. have demonstrated that SCCO_2 can be used as an effective medium for HF etching of SiGe native oxides [101].

HF/ H_2O in SCCO_2 has been shown to etch tetraethyl orthosilicate (TEOS) and release silicon cantilevers, held in place for fabrication by SiO_x , without stiction or residues. Furthermore, etching rates are dependent on the HF concentration and temperature; however, the water concentration in the SCCO_2 (maintaining the same HF/ H_2O ratio) and the pressure had very limited effect on the etching rate [102]. The addition of alcohols to this process decreases the TEOS etching rate, but shows enhanced selectivity for some materials such as silicon nitride.

8.3.7 Pore Sealing

Metal intrusion into porous low-k films during deposition limits device performance and reliability. Metal species penetration into pores can increase the effective dielectric constant (k_{eff}) and current leakage, and lower the breakdown voltage. The addition of SiN or SiCN barrier layers to inhibit Cu diffusion also will increase the k_{eff} of the dielectric stack structure. Recently, methods of sealing pores to protect against metal intrusion have been demonstrated by techniques such as deposition of thin low-k polymeric films [103], plasma treatments [104], and surface functionalization with large pendant groups using SCCO_2 [83,85]. In each case, the sealing layer should extend into the pore perimeters. To extend the effective electron cloud of the sealing layer, large alkyl or aromatic silylating agents are reacted with surface silanol groups.

Two different approaches are employed: trialkylsilane of increasing alkyl group sizes and monoalkylsilanes. Trialkylsilanes do not show significant improvement of surface hydrophobicity over smaller reagents, such as HMDS, indicating that steric issues limit the number of silanol groups that are available for reaction [83]. Recent work has shown that surface functionalization and the layer thicknesses can be adjusted to best control intrusion of chemical species into the porous low-k [105].

Initial efforts to examine the effectiveness of silylating agents as chemical sealants employ SCCO_2 /HMDS-treated MSQ in the presence of a metal barrier precursor, i.e., tetrakis-dimethylamido titanium (TDMAT) [77]. SEM images indicated that the Ti reaction products do not penetrate into the low-k film with the use of the chemical sealant, whereas Ti species are found in unsealed MSQ, as shown in Fig. 8.3-35. Furthermore, via-chain electrical shorts showed considerable improvement after silylating agents, suggesting the prevention of metal penetration sealed the MSQ. Most recently,

**Titanium
Reaction Products**

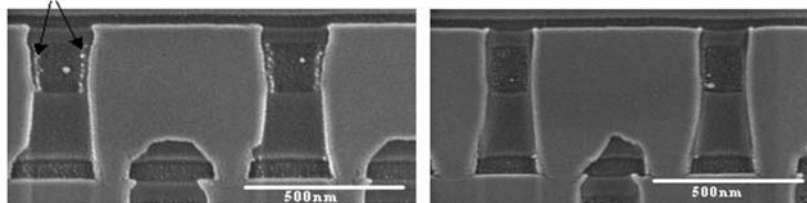


FIGURE 8.3-35 Titanium penetration into a porous MSQ film after reaction with TDMAT (left). This intrusion is prevented by sealing with SCCO_2 /HMDS (right) [77]. *Used with permission from Richard F. Reidy.*

high-resolution TEM results showed that ruthenium (Ru) can be prevented from penetrating into a plasma-damaged MSQ film after exposure to HMDS [106]. Monoalkylchlorosilanes show promising results with respect to increasing the contact angles, thus increasing the hydrophobicity, as a function of the alkyl tail length [85].

8.3.8 Supercritical Fluid Drying and Porogen Removal

To form controlled porosity in low-k dielectrics several methods have been examined, including a range of templating techniques [107,108]. Work by researchers at the University of Missouri, Texas Tech University, and IBM have focused on the use of organic pore templates that can be decomposed thermally and extracted using SCCO_2 . Both open and closed pore structures were observed with pore sizes between 1 and 2 nm. Porogen loading can range from 25 to 55 wt% resulting in 27%–62% porosity. Issues remain with these methods in terms of complete porogen removal, although supercritical extraction appears to be superior to thermal decomposition alone. Lubguban et al. report that use of cosolvents (methyl ethyl ketone and tetrahydrofuran) can remove porogens in templated films, producing low-k films with dielectric constants of 1.61 (open pore) and 2.3 (closed pore) [109].

SCCO_2 processes are shown to dry high-aspect ratio resist lines without collapse and to remove H_2O and reaction by-products from porous low-k films. In 2000 Goldfarb et al. and Namatsu described the use of SCCO_2 with solvents and surfactants to dry water-rinsed photoresists [110,111]. More recently, a range of options using other surfactants and cosolvents to limit the capillary forces during drying is used in SCCO_2 [112,113].

Several groups developed photoresists that are deposited, and later stripped in SCCO_2 [114–117]. SCF are employed to develop resists using CO_2 -compatible salts [118].

8.4 SUMMARY OF SUPERCRITICAL FLUIDS

SCF are a potential processing technology for the semiconductor industry. In this chapter, techniques are outlined that use SCCO₂ to strip photoresist, remove postetch residues, remediate low-k films, and seal pores. These SCCO₂ processes are shown to have advantages over baseline processes. Supercritical methods are quite effective at removing etch-hardened 193-nm photoresist on low-k films and require no postclean drying steps as the SCCO₂ also dries the film. SCCO₂ processes are shown to dry high-aspect ratio resist lines without collapse and to remove H₂O and reaction by-products from porous low-k films.

During etching and plasma stripping of porous low-k films, carbon species are depleted from the surface. Supercritical silylation reactions demonstrate effective repair of the dielectric constant through the removal of moisture and replacement of the lost hydrophobic carbon-containing moieties. In addition, SCCO₂-based silylation reactions prevent the loss of the sidewall film that has been exposed to the plasma process that are susceptible to removal during subsequent wet etches, plus these processes inhibit the penetration of metallic species. Environmental advantages of the technology include the elimination or reduction of the use of H₂O and other aqueous chemicals during cleaning.

In addition to cleaning, there are several areas where supercritical fluid technology shows great promise. Device-quality metal and dielectric films have been successfully deposited using SCF. Porogens for templated low-k films are efficiently removed using SCCO₂. Some efforts indicate that supercritical processes can effectively remove particles from wafer surfaces.

Adoption will likely occur when traditional wet cleaning becomes yield limiting as a result of high-aspect ratio structures that cannot be reproducibly cleaned and dried. The introduction of porous ultralow-k films may also provide a driving force to implement SCCO₂ processes. Greater porosity increases the need to have a nondestructive means of removing contamination that may limit performance and reliability. Overall, supercritical fluid technology offers a number of potential solutions to current and future challenges in semiconductor manufacturing.

However, there remain several key issues with respect to the adoption of SCCO₂ processing in semiconductor manufacturing. Although benefits over baseline stripping and cleaning processes have been demonstrated with SCCO₂, there is some reluctance among IC manufacturers to introduce SCCO₂ processes and equipment manufacturers to develop high-volume production-worthy tools. Significant cost savings and yield improvements must be demonstrated. In addition, for SCCO₂ implementation, more work on eliminating contamination from equipment and improving reliability and process uniformity is needed.

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